

Turning points in primary care disease trajectories

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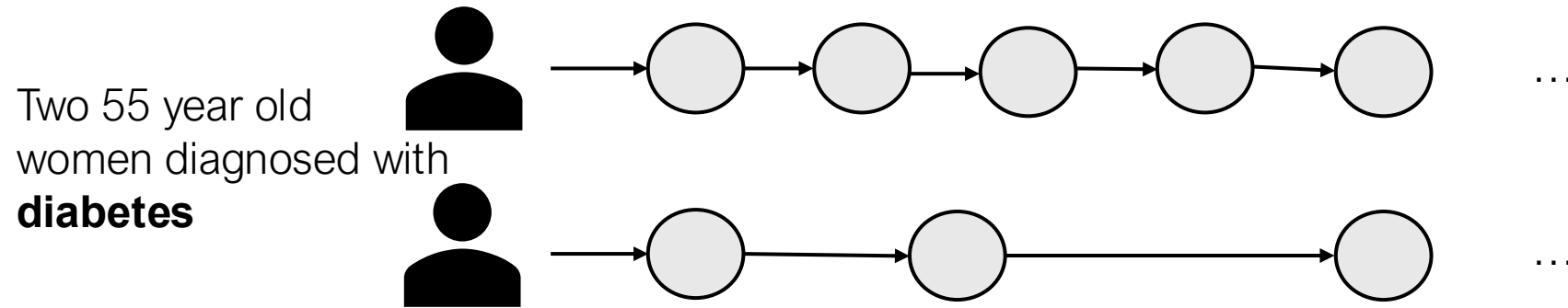
July 3, 2026

Joint work with Nathaniel Hendrix, Diego Guerrero, Laust Mortensen and David Rehkopf

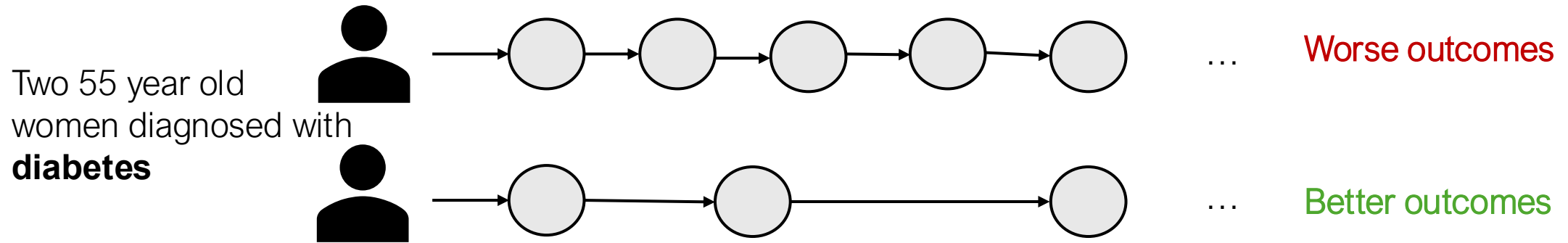
Disclosure of speaker's interests

No (potential) conflict of interests

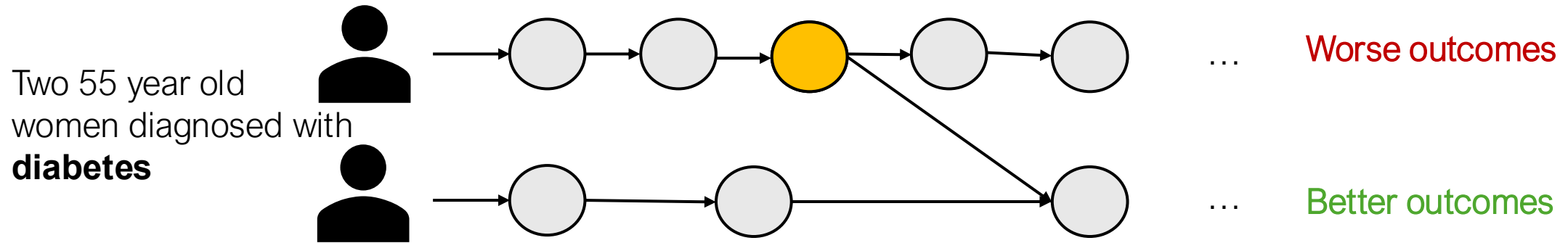
For more information see <https://www.iagg2026.org/resources/uploads/sites/137/2026/01/Format-of-disclosure-slide-for-speakers-ENG.pdf>



Each person has a health **trajectory**



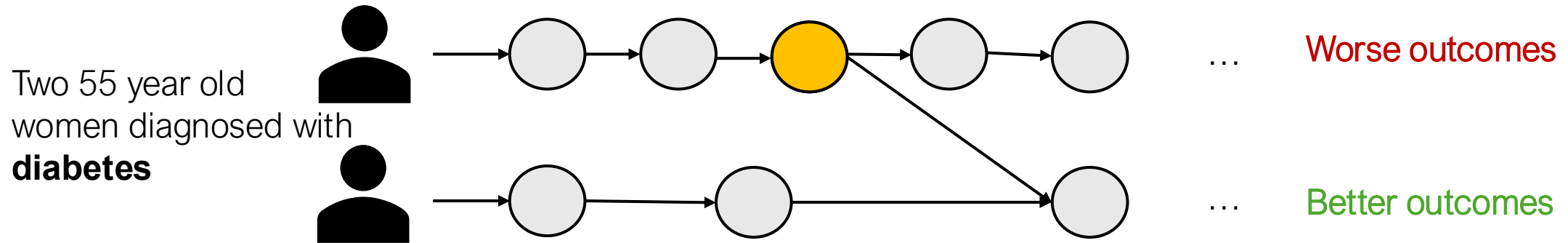
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Are there diagnoses that persistently redirect a patient's trajectory to one with better outcomes?

The idea of “**turning points**” is theorized but hard to operationalize
But really important for intervention



Each person has a health **trajectory**

Are there diagnoses that persistently redirect a patient's trajectory to one with better outcomes?

A lot of prior work uses European registry data

Temporal disease trajectories condensed from population-wide registry data covering **DK** 6.2 million patients

Anders Boeck Jensen^{1,2}, Pope L. Moseley^{2,3}, Tudor I. Oprea^{1,3,4}, Sabrina Gade Ellesøe², Robert Eriksson^{1,2}, Henriette Schmock⁵, Peter Bødstrup Jensen², Lars Juhl Jensen² & Søren Brunak^{1,2}

2014

Disease trajectory browser for exploring temporal, population-wide disease progression patterns in 7.2 million Danish patients

Troels Siggaard¹, Roc Reguant¹, Isabella F. Jørgensen¹, Amalie D. Haue¹, Mette Lademann¹, Alejandro Aguayo-Orozco¹, Jessica X. Hjaltelin¹, Anders Boeck Jensen^{1,2}, Karina Banasik¹ & Søren Brunak¹

DK
2020

Unraveling cradle-to-grave disease trajectories from multilayer comorbidity networks **AT**

 Check for updates

Elma Dervic^{1,2,3}, Johannes Sorger¹, Liuhuaying Yang¹, Michael Leutner⁴, Alexander Kautzky⁵, Stefan Thumer^{1,3,6}, Alexandra Kautzky-Willer^{4,7} & Peter Klimek^{1,2,3}

2024

Mapping multimorbidity progress among 190 diseases

Learning prevalent patterns of co-morbidities in multichronic patients using population-based healthcare data

Chiara Seghieri¹, Costanza Tortù¹, Domenico Tricò² & Simone Leonetti^{1,3,4}

GB

Shasha Han^{1,2,3}, Sairan Li¹, Yunhaonan Yang⁴, Lihong Liu⁵, Libing Ma⁶, Zhiwei Leng⁷, Frances S. Mair⁸, Christopher R. Butler^{9,10}, Bruno Pereira Nunes^{11,12}, J. Jaime Miranda^{13,14}, Weizhong Yang^{1,2,3}, Ruitai Shao^{1,2,3} & Chen Wang^{1,2,3,15}

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2024

Prior work often uses hospital data, often from a single care system, or for a **single disease**. And focuses on predicting outcomes, without understanding where their paths separate.

Primary care is uniquely rich!

American Family Cohort data



7.9 million patients



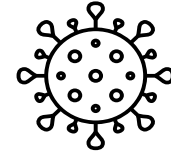
1,331 clinics in the U.S.



4,573 providers



2017-2024



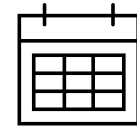
Diagnoses



Prescriptions



Labs



Visits

This is the setting where
divergence in trajectories is likely
most visible

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7.9 million patients



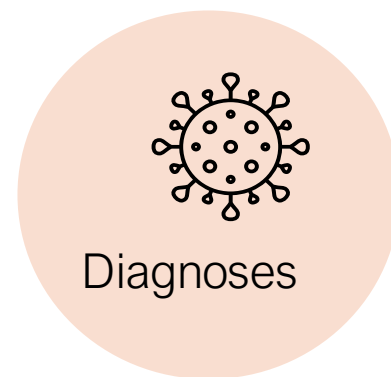
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Would like to incorporate the
other info eventually!

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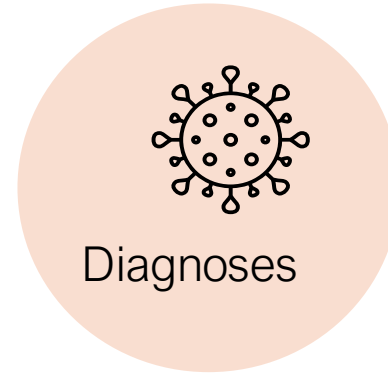
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Prescriptions

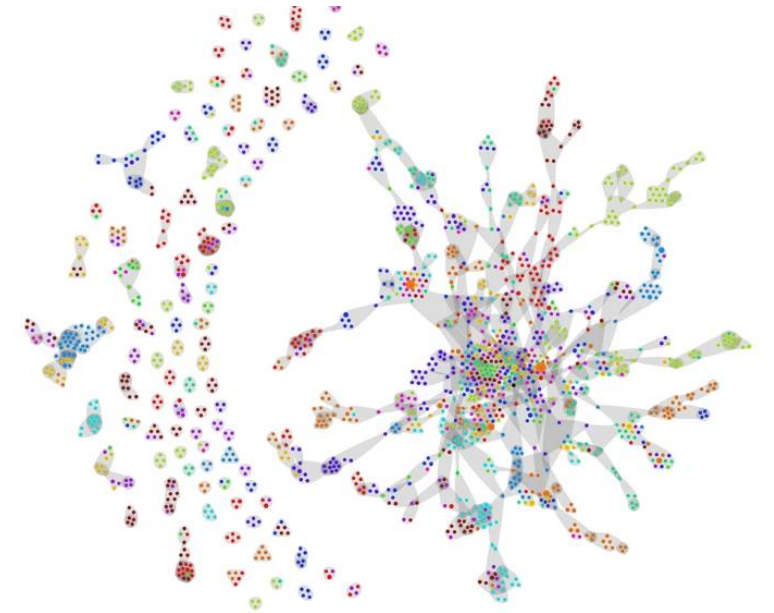


Visits

Use a **network**, a way of representing complex data

Maps how **diseases cluster and progress together**

Let's the strongest relationships emerge

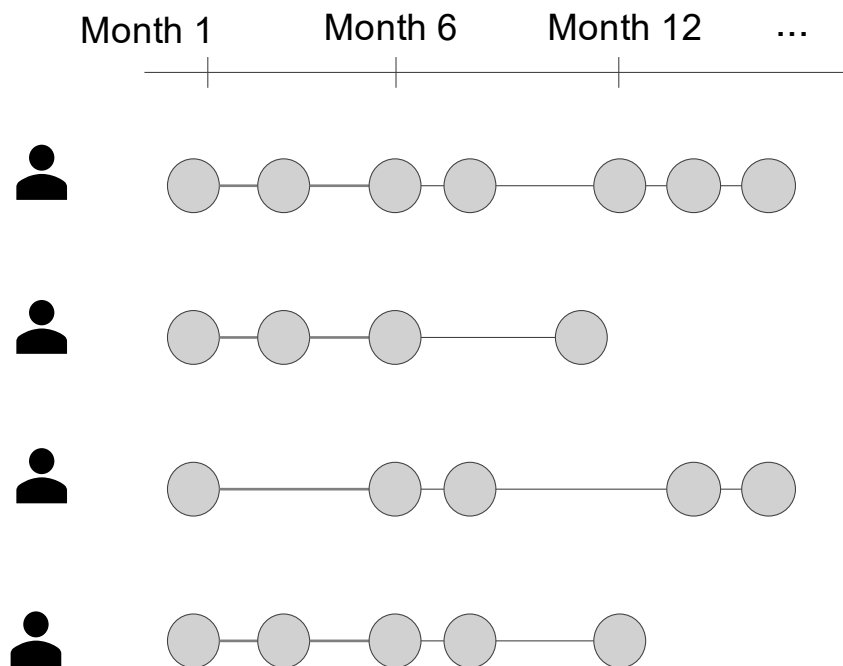


What is the disease trajectory landscape in U.S. primary care?

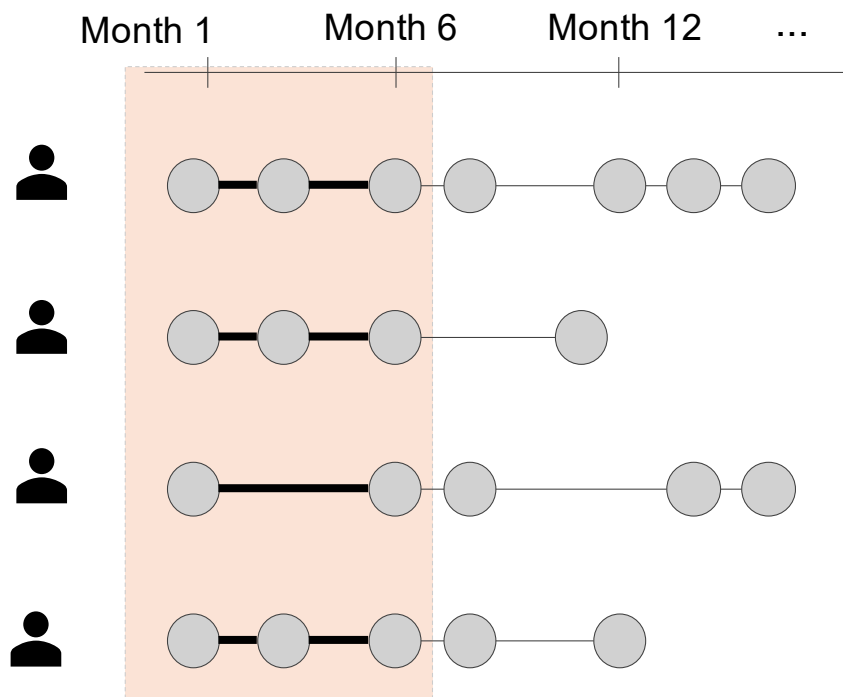
How can we operationalize turning points?

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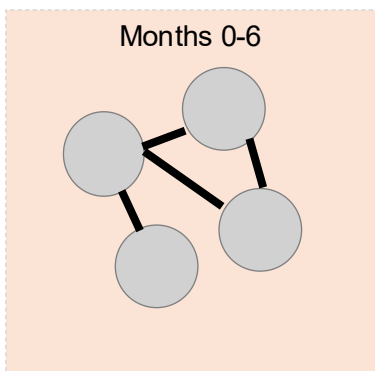
**Each person has a
sequence of diagnoses
over 5 years**

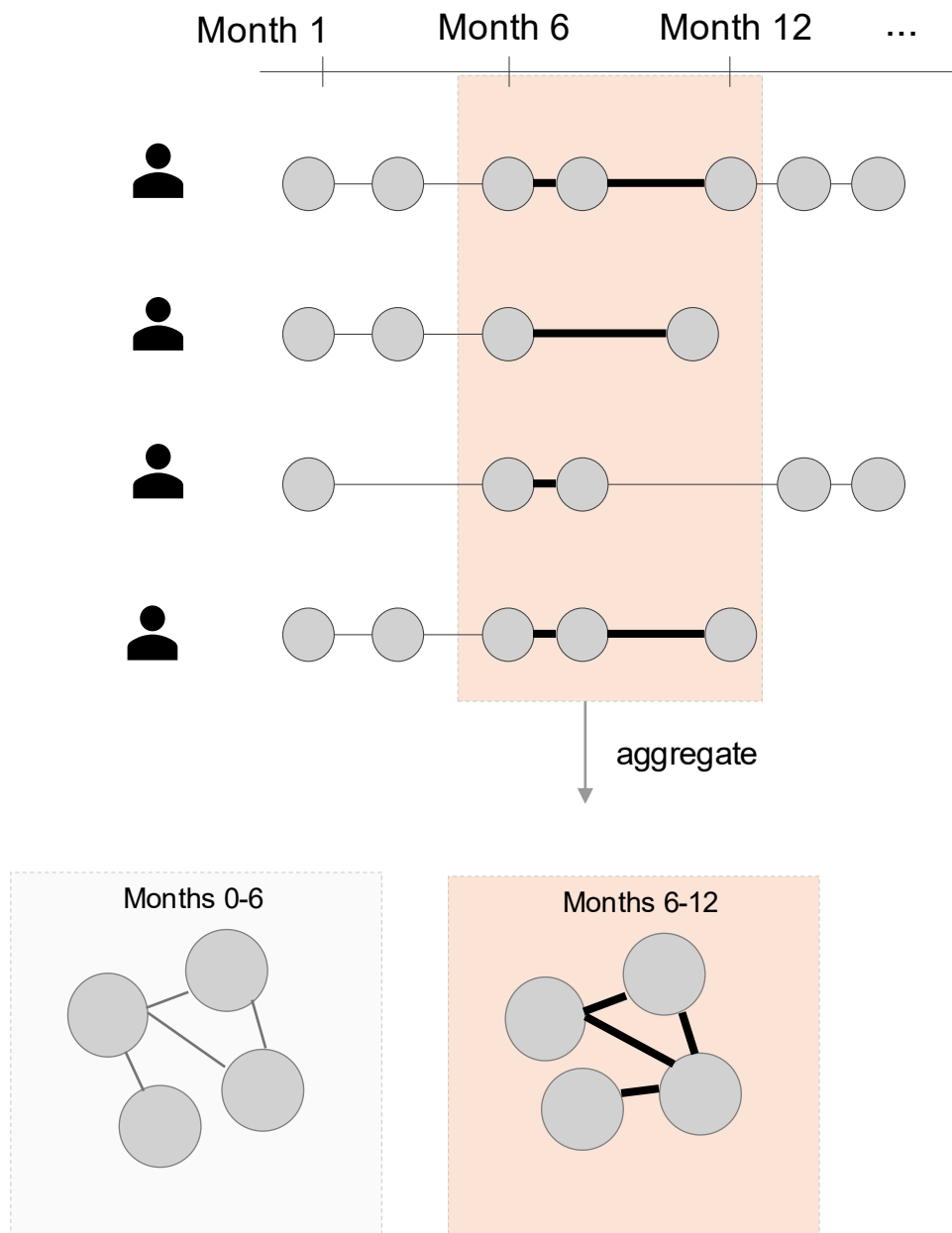


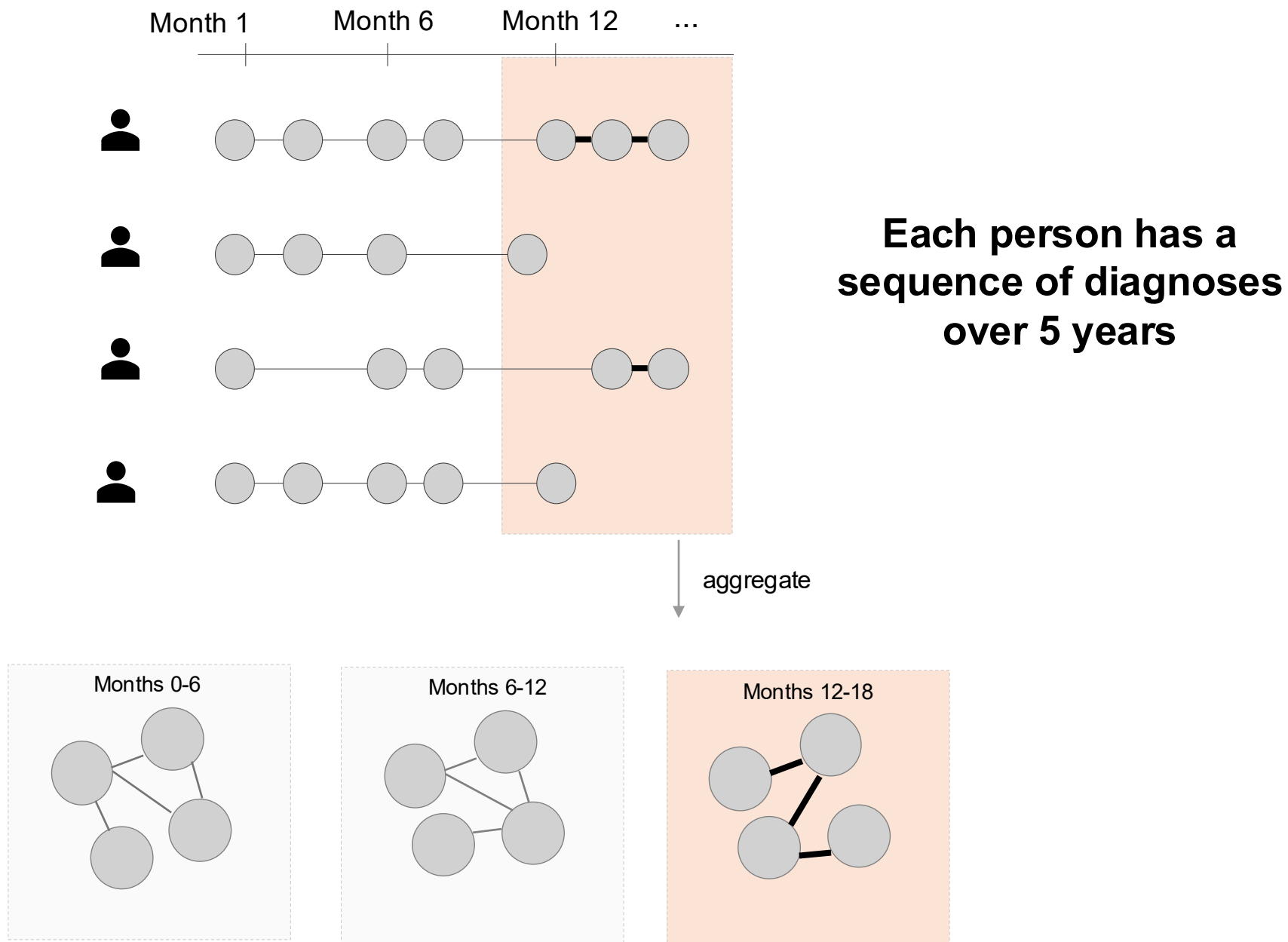
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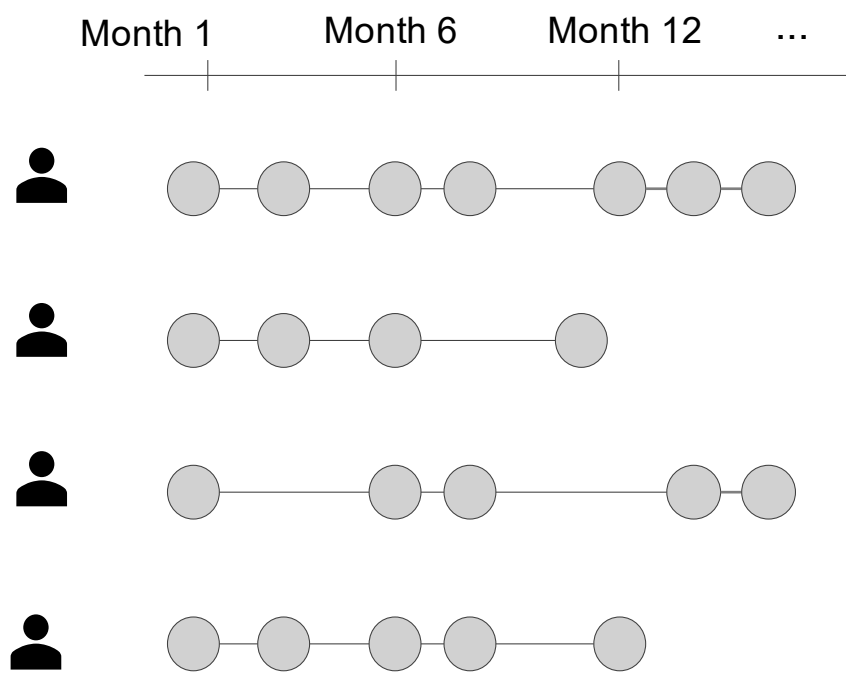
aggregate

An **edge** between
two diseases
means that pair
co-occurs at a
greater rate than
expected in the
population





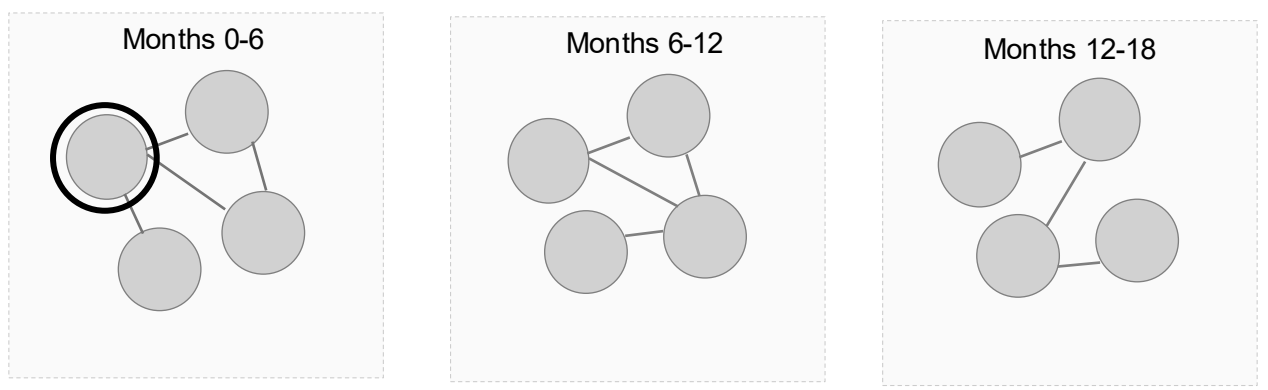




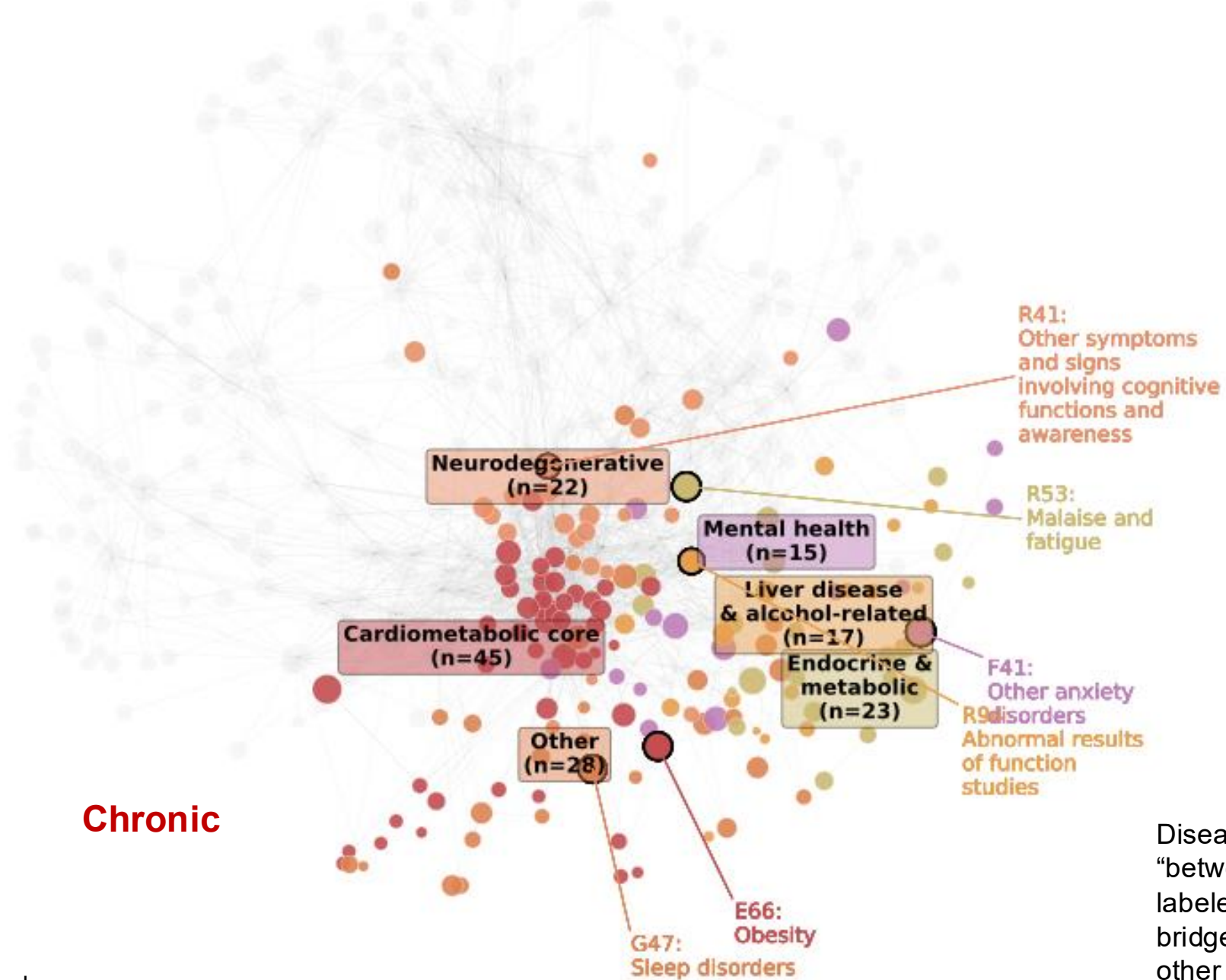
Each person has a sequence of diagnoses over 5 years

aggregate

High “betweenness” centrality means that disease is a **bridge** between many other diseases

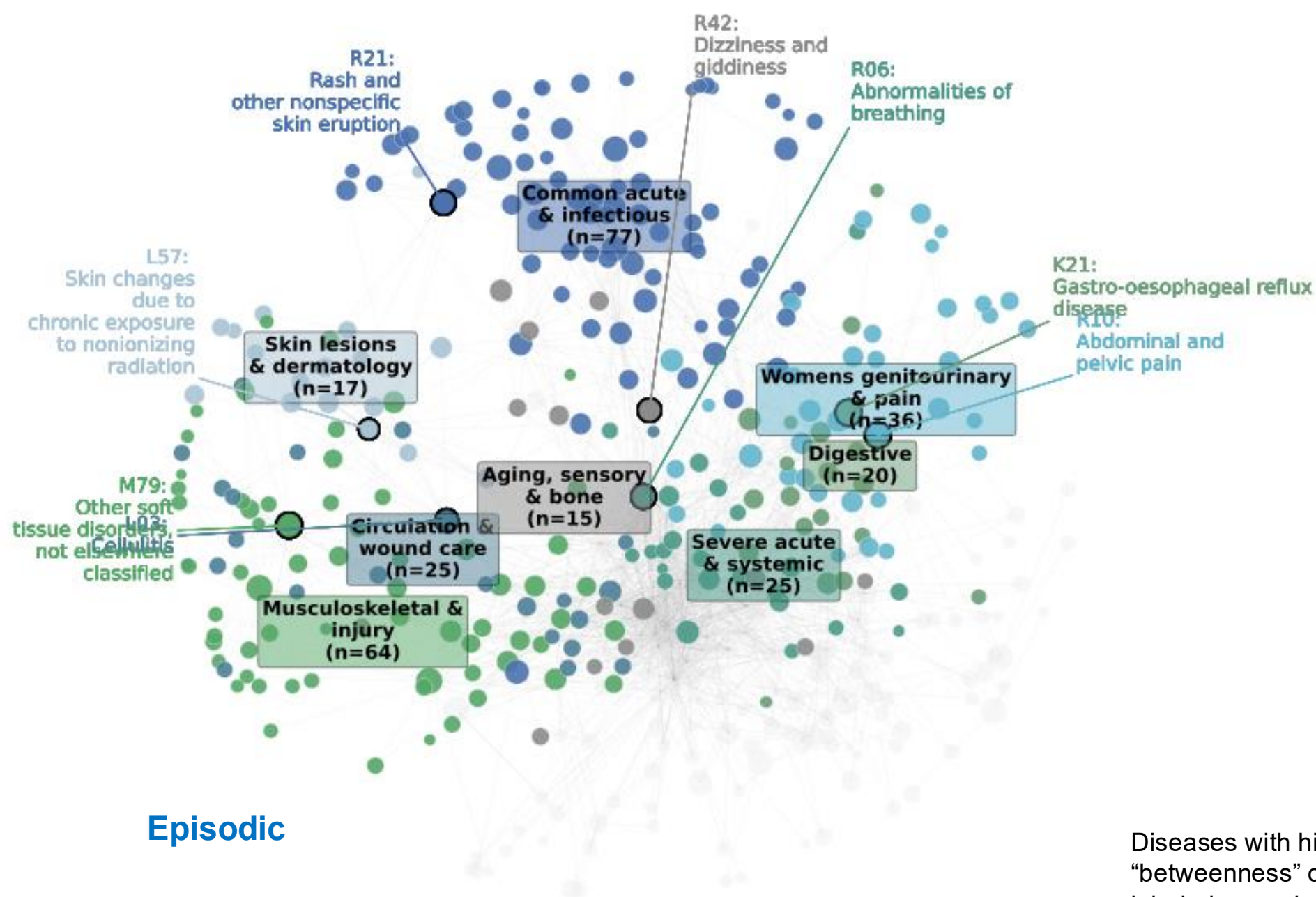


When we aggregate edges across time, two main disease clusters emerge

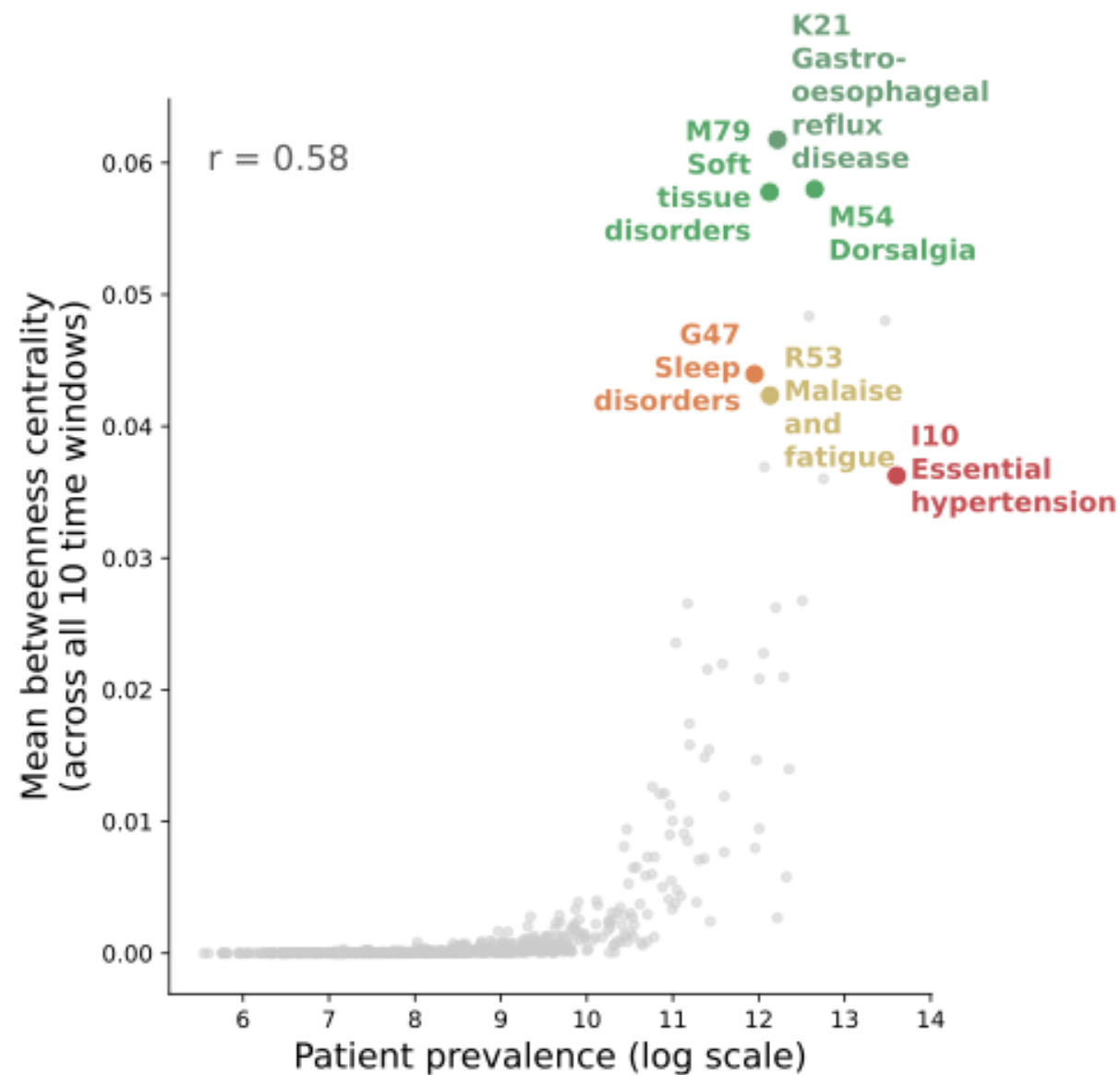


Chronic

Diseases with high “betweenness” centrality are labeled, meaning they are bridge nodes between many other diseases



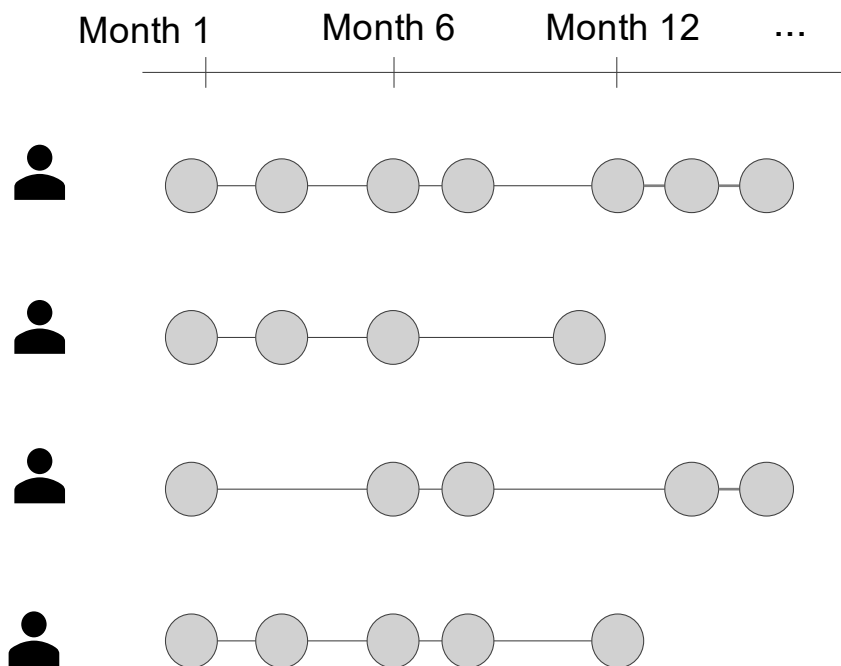
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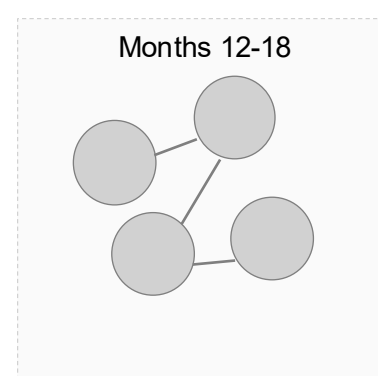
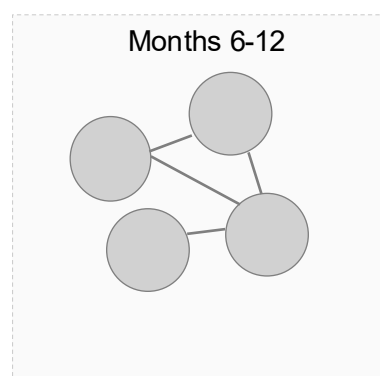
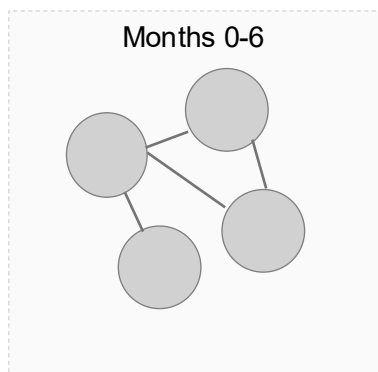
A small set of diseases connect groups of diseases (bridge nodes)

They tend to be prevalent but higher prevalence $\neq$ higher betweenness

The highest ones are episodic

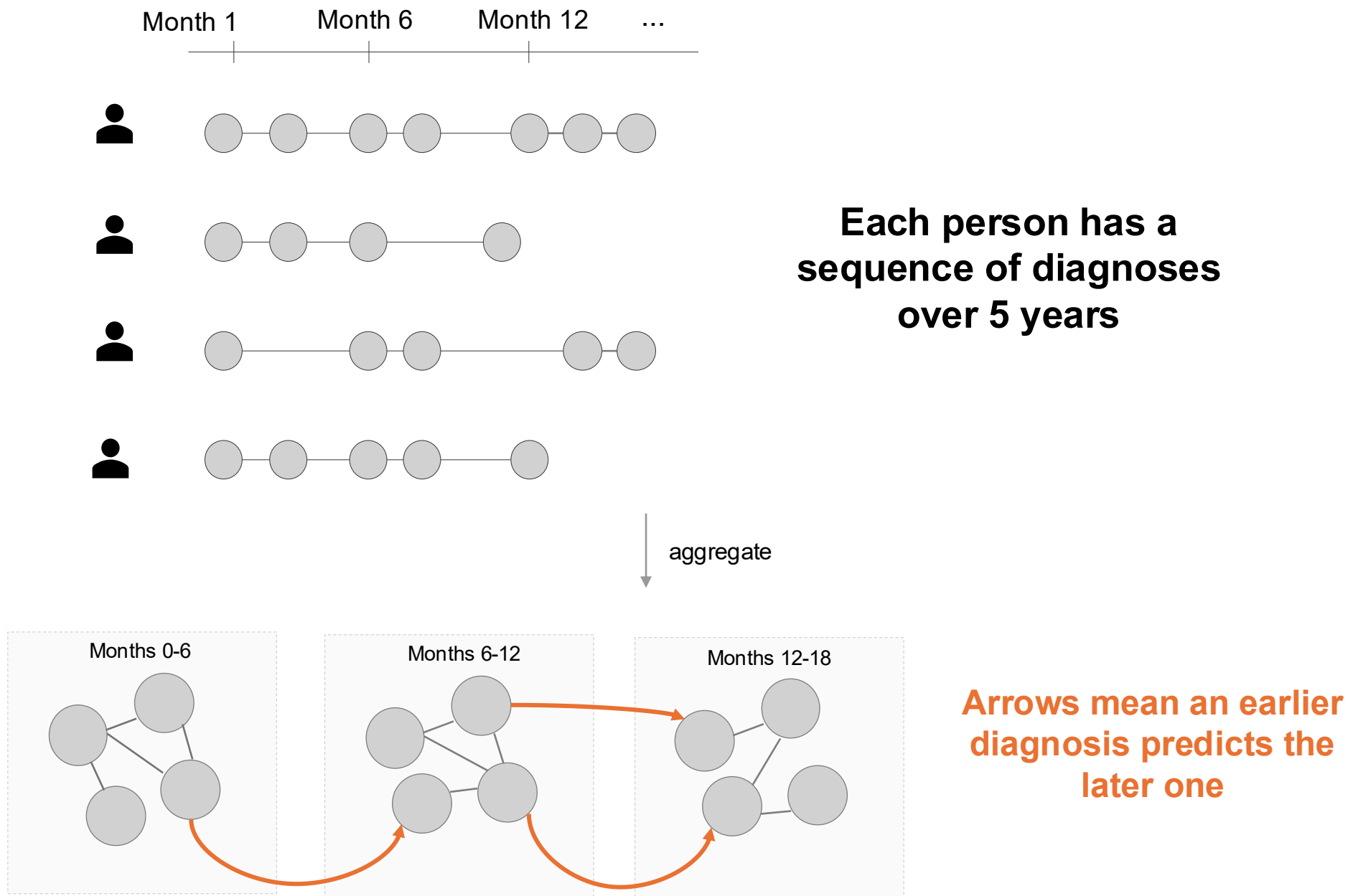


↓ aggregate

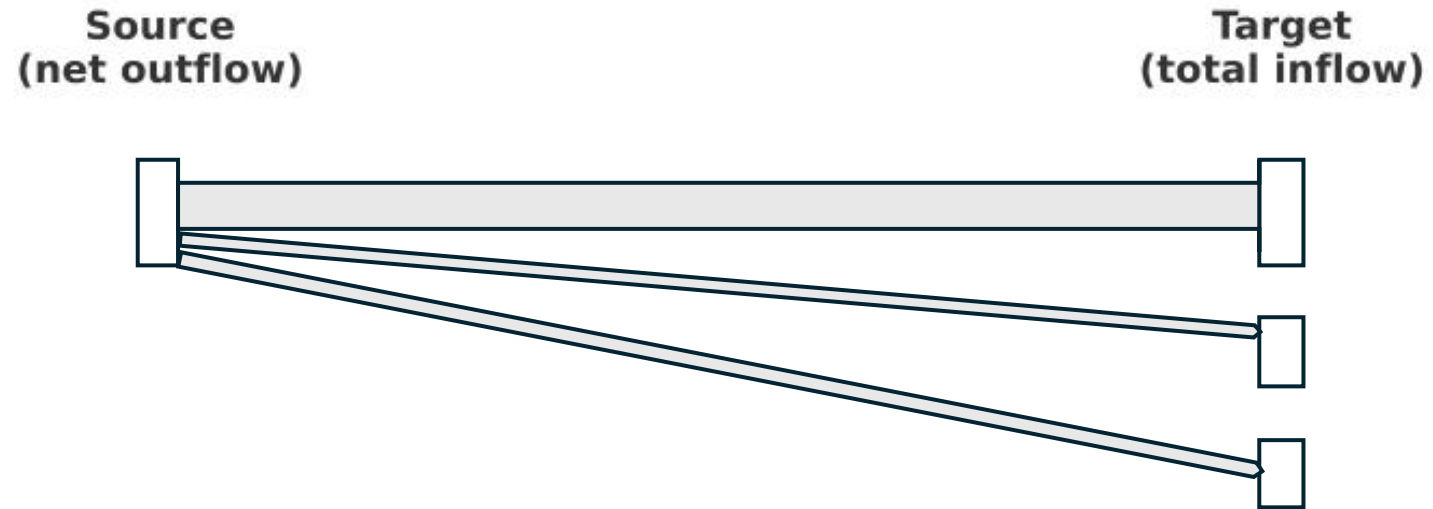


These edges tell us which diseases cluster together (comorbidity)

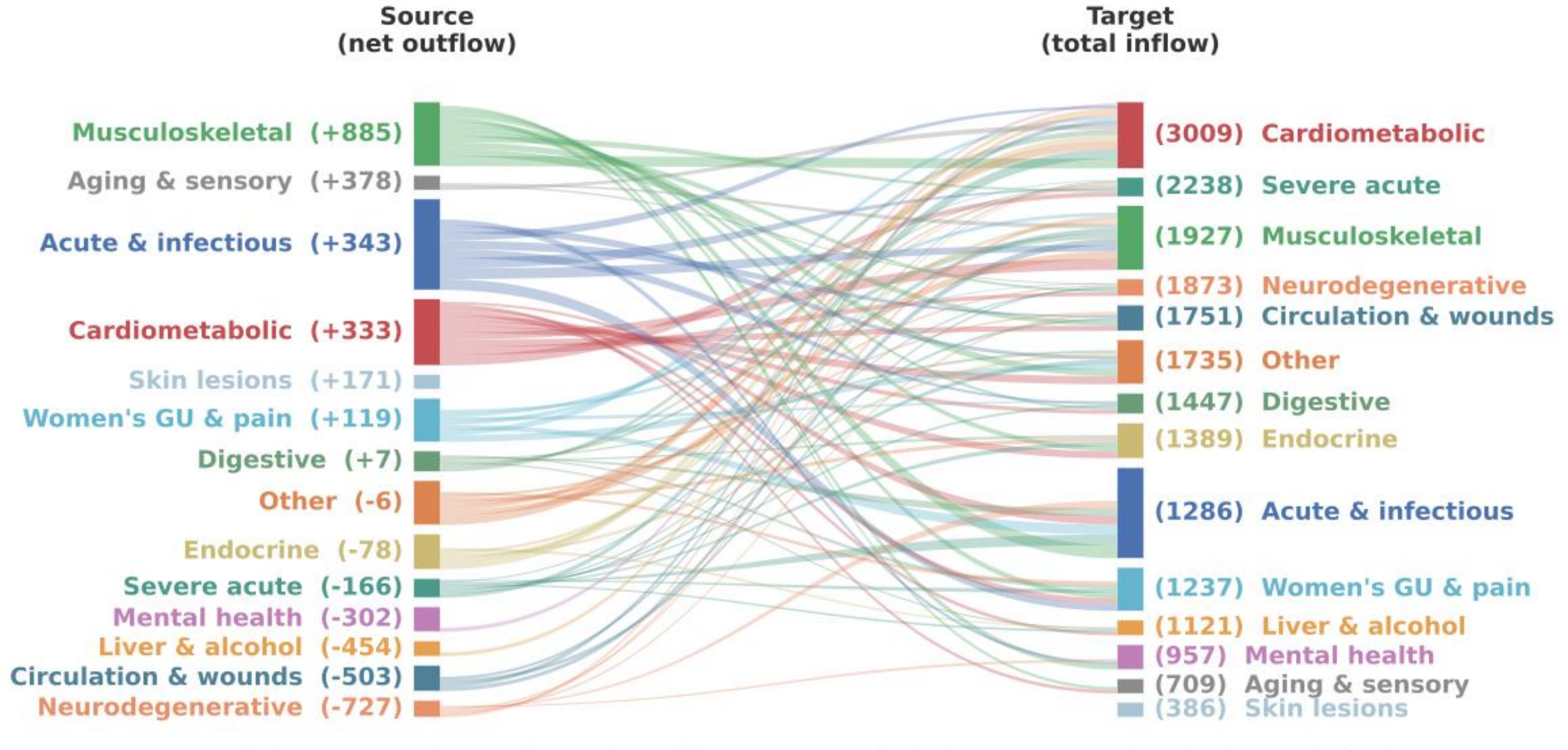
But the order the diseases were diagnosed matters too



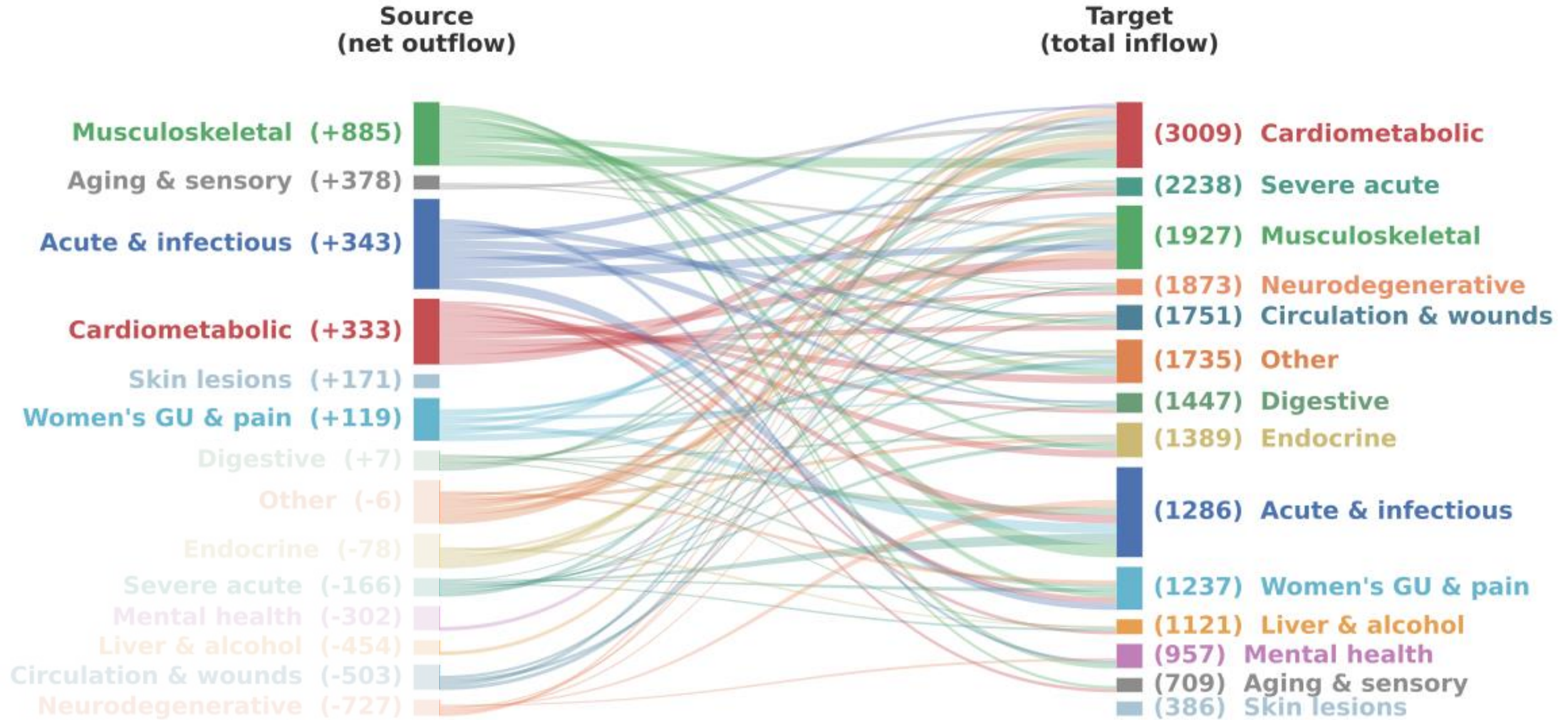
Which diseases are upstream versus downstream?



Which diseases are upstream versus downstream?

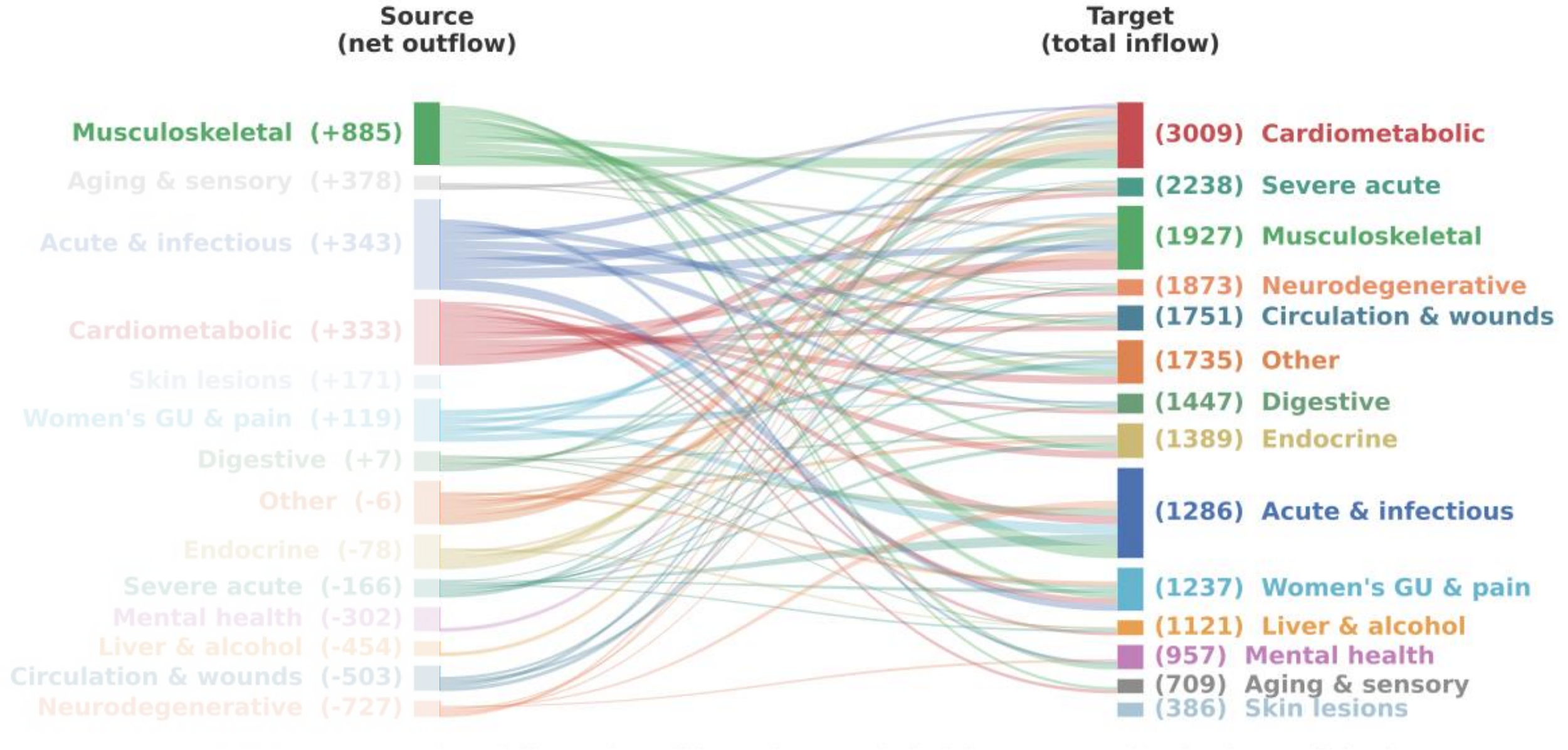


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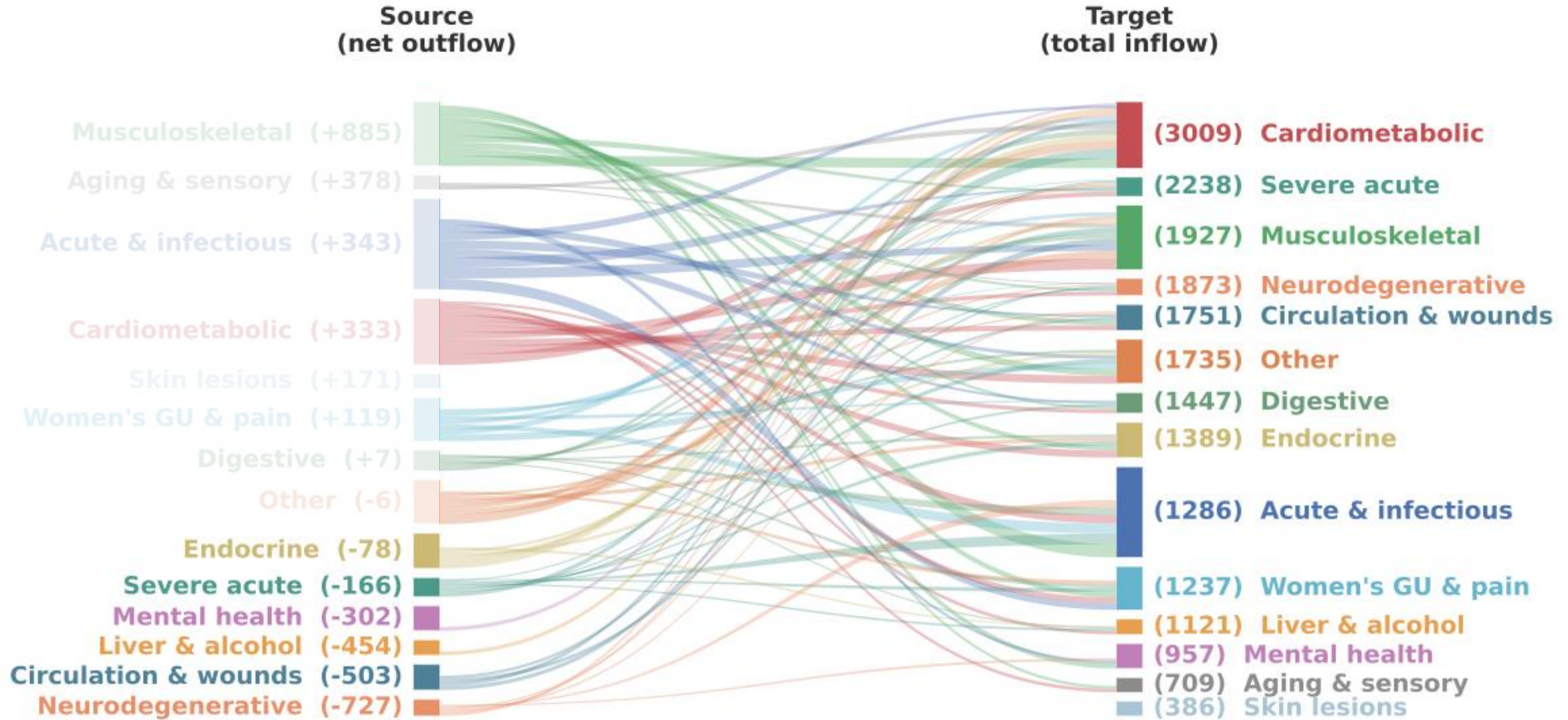


These disease clusters are **net sources**, meaning they predict more diseases upstream than they are predicted downstream

Which diseases are upstream versus downstream?



Which diseases are upstream versus downstream?



These disease clusters are **net sinks**, meaning they are more often predicted downstream

What is the disease trajectory landscape in U.S. primary care?

Population

*Tells us which diseases connect the landscape, how chronic disease progresses
Doesn't tell us why our two women end up in different places*

What is the disease trajectory landscape in U.S. primary care?

How can we operationalize turning points?

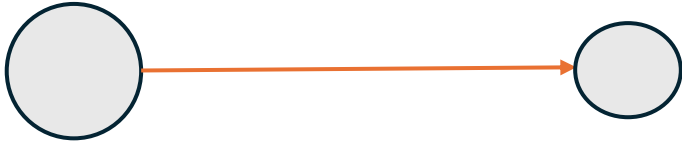
Population



Individual

Counterfactual design

T2DM diagnosis

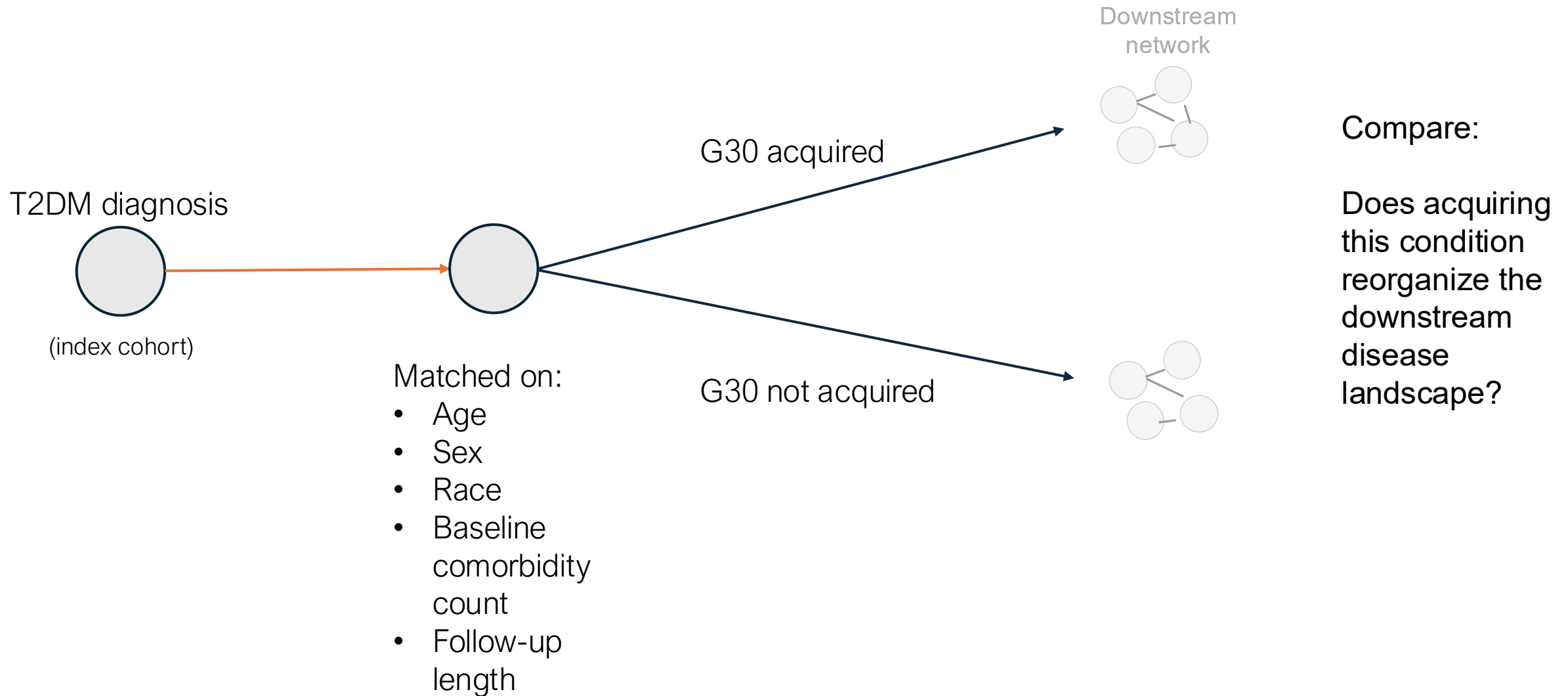


(index cohort)

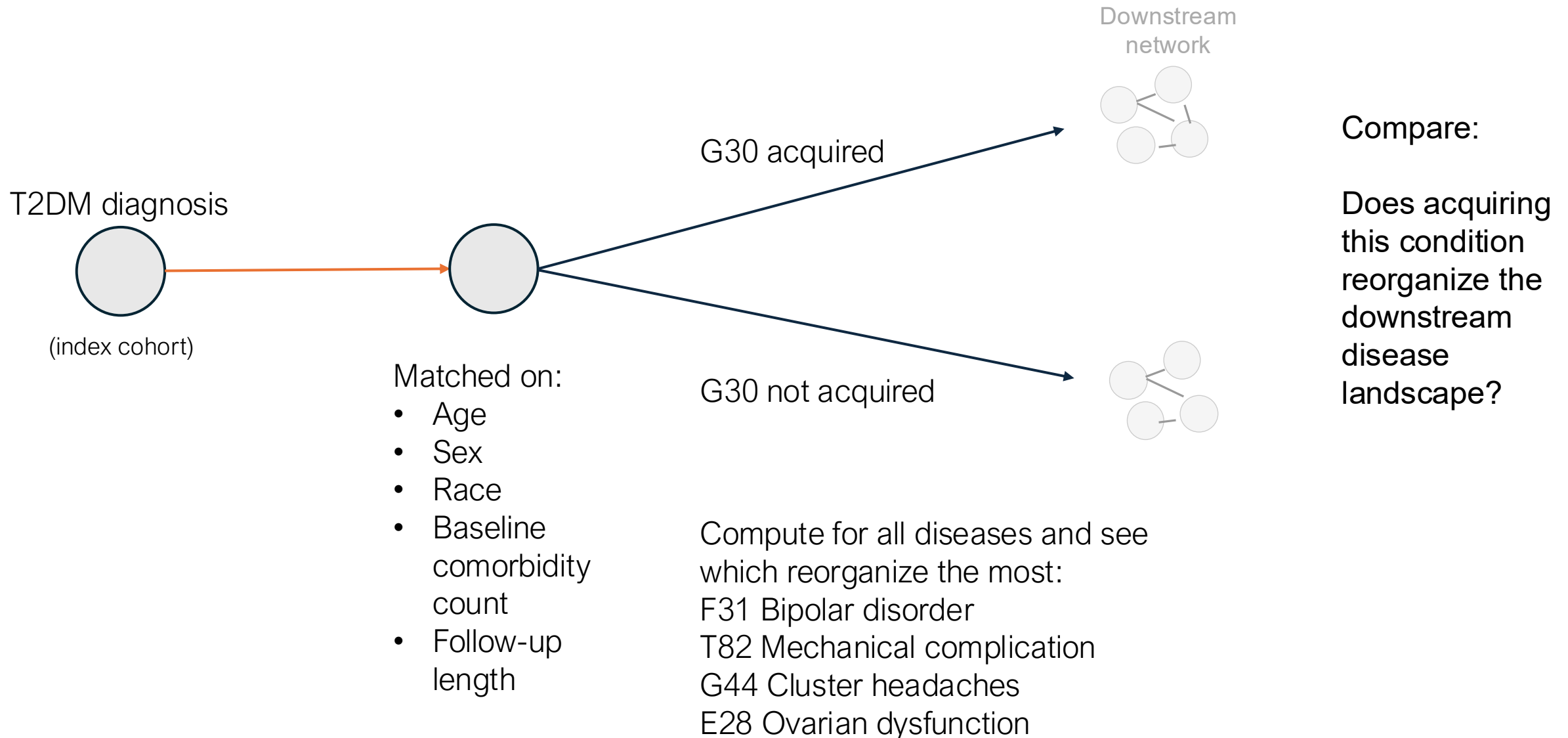
Matched on:

- Age
- Sex
- Race
- Baseline
comorbidity
count
- Follow-up
length

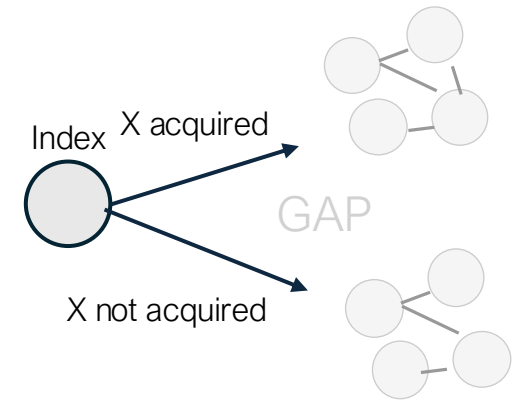
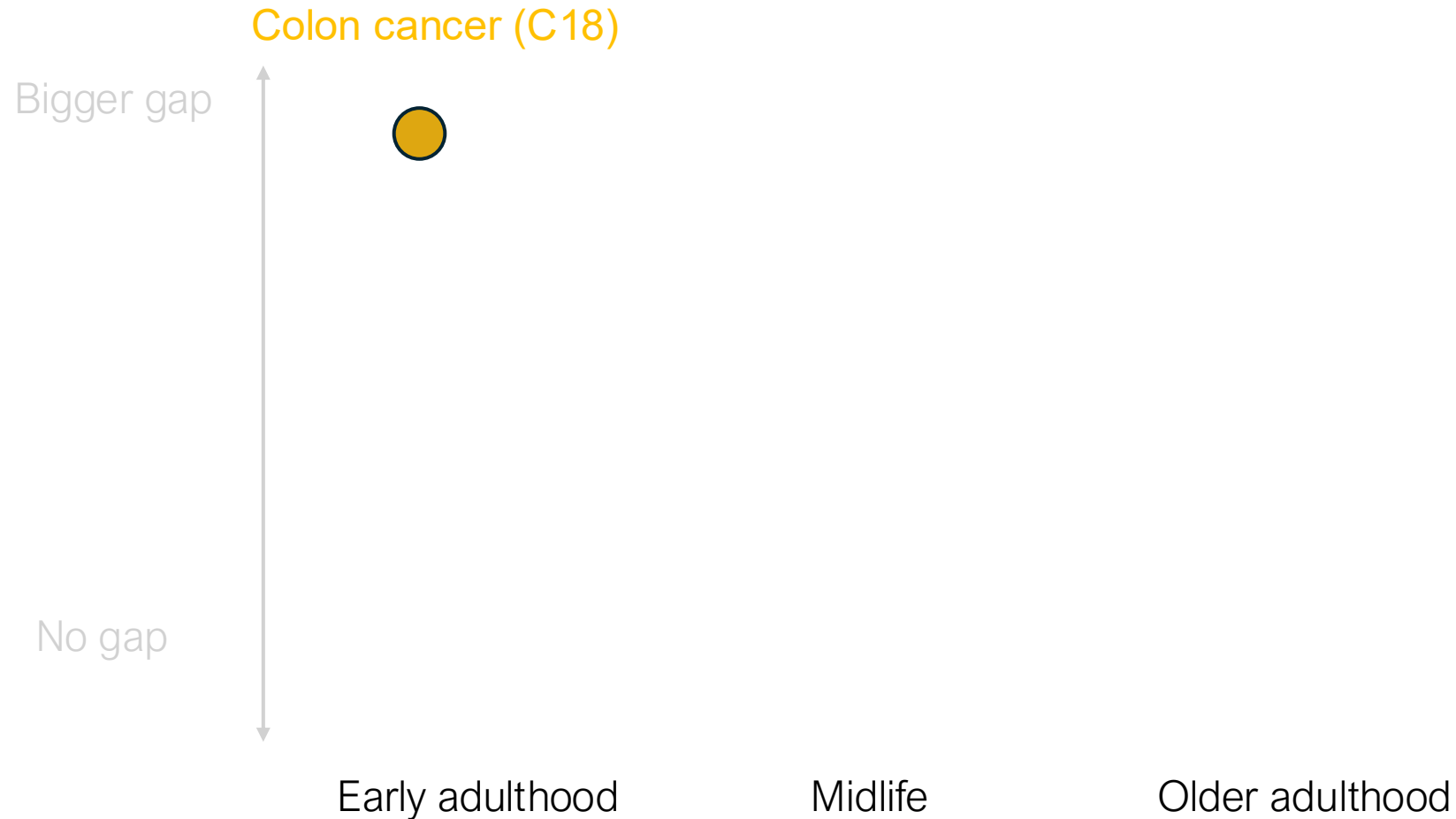
Counterfactual design



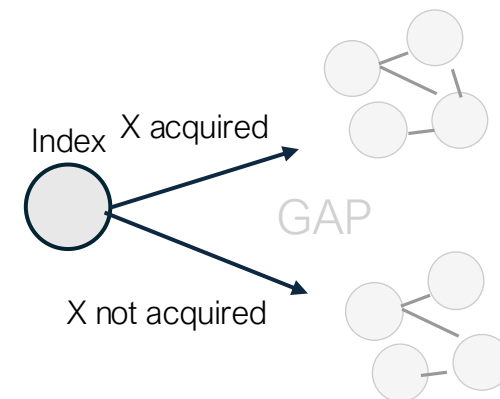
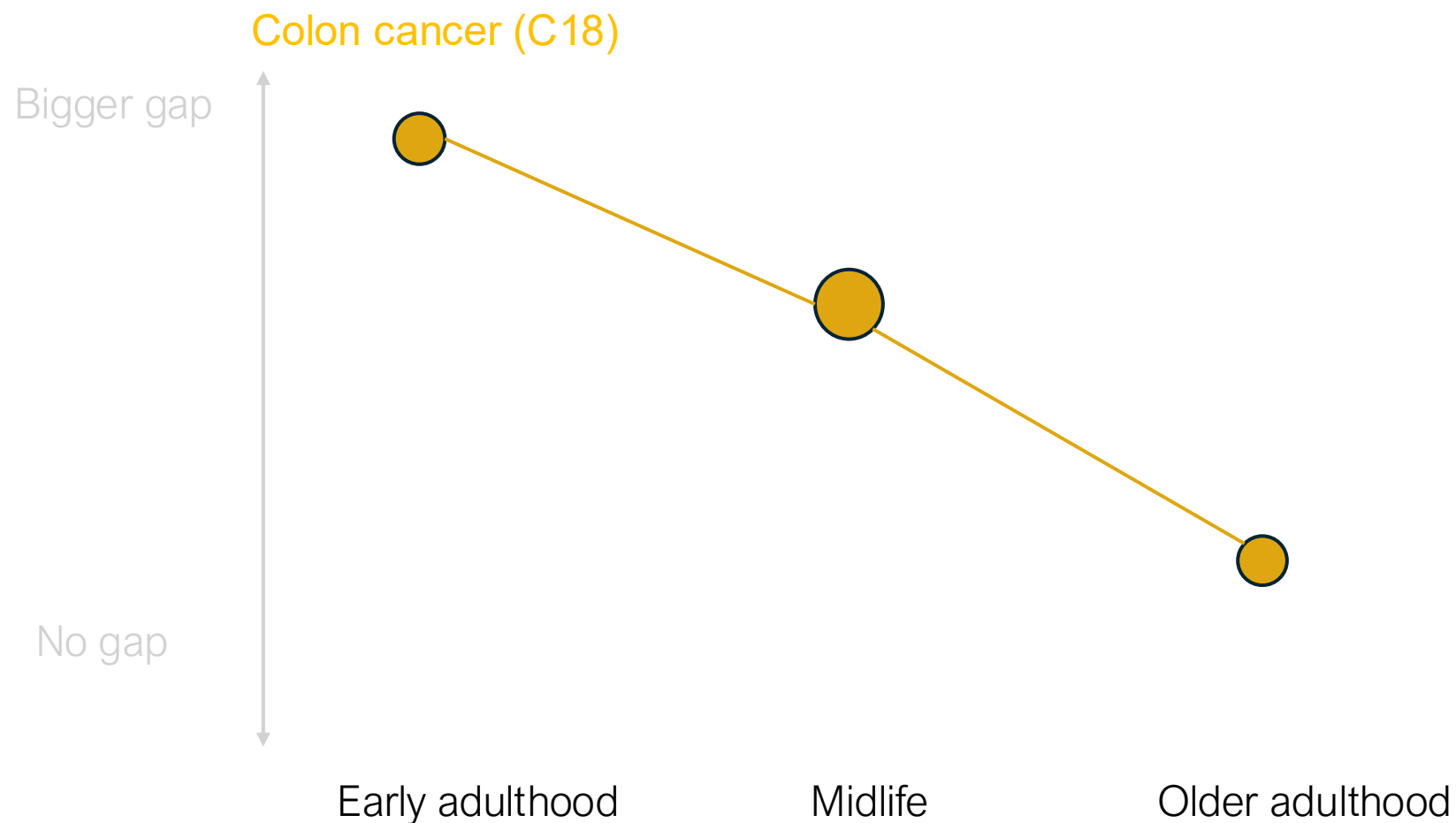
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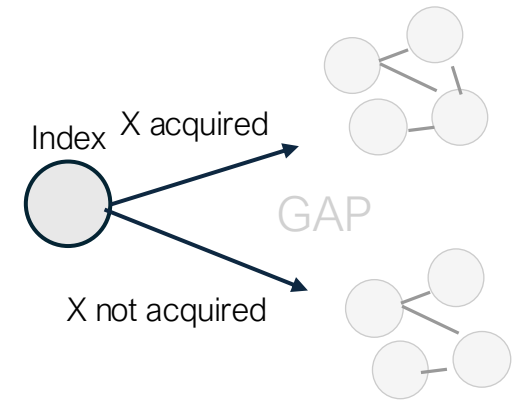
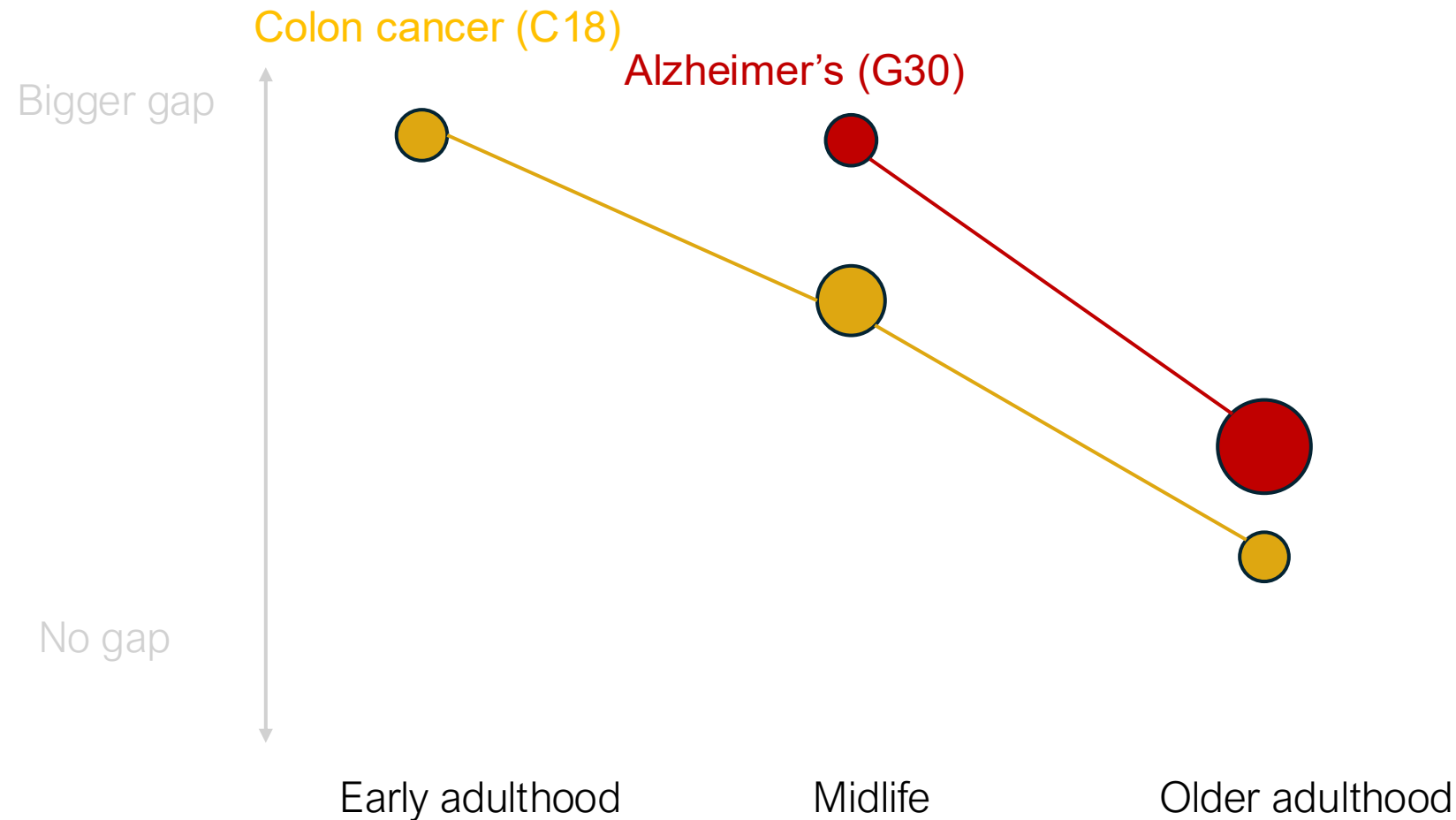
But when we split by age...



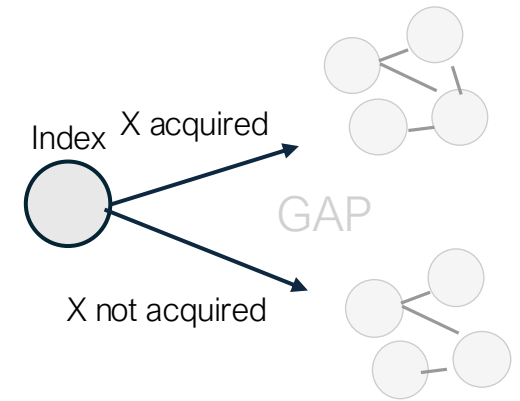
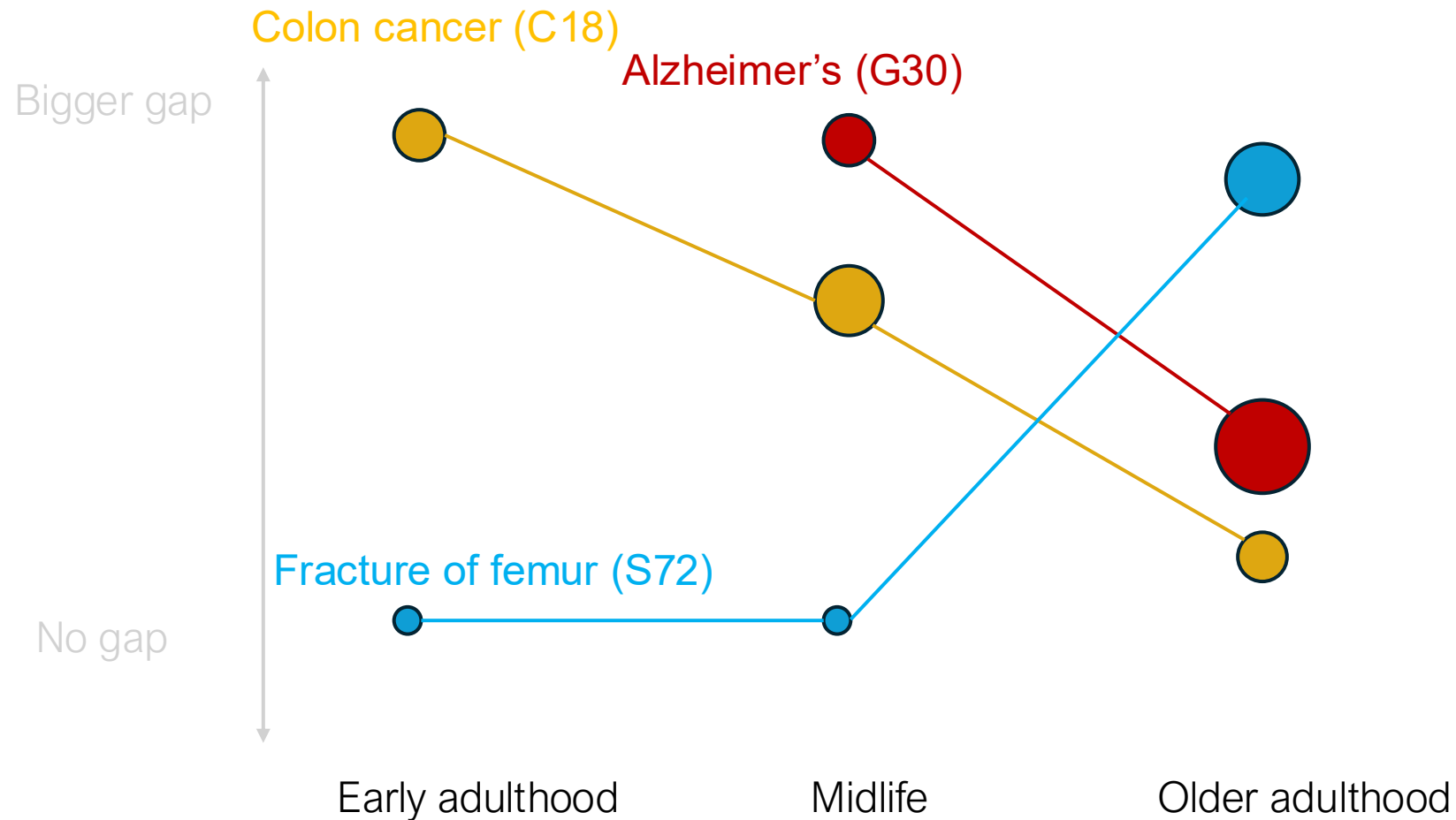
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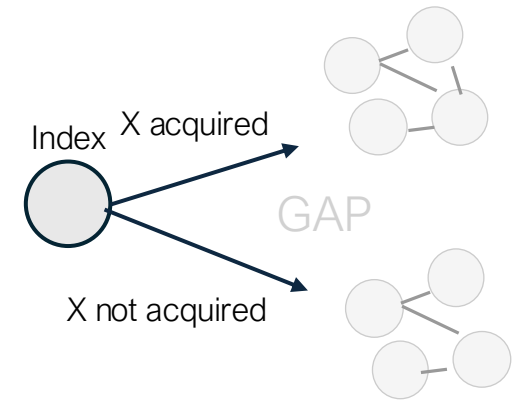
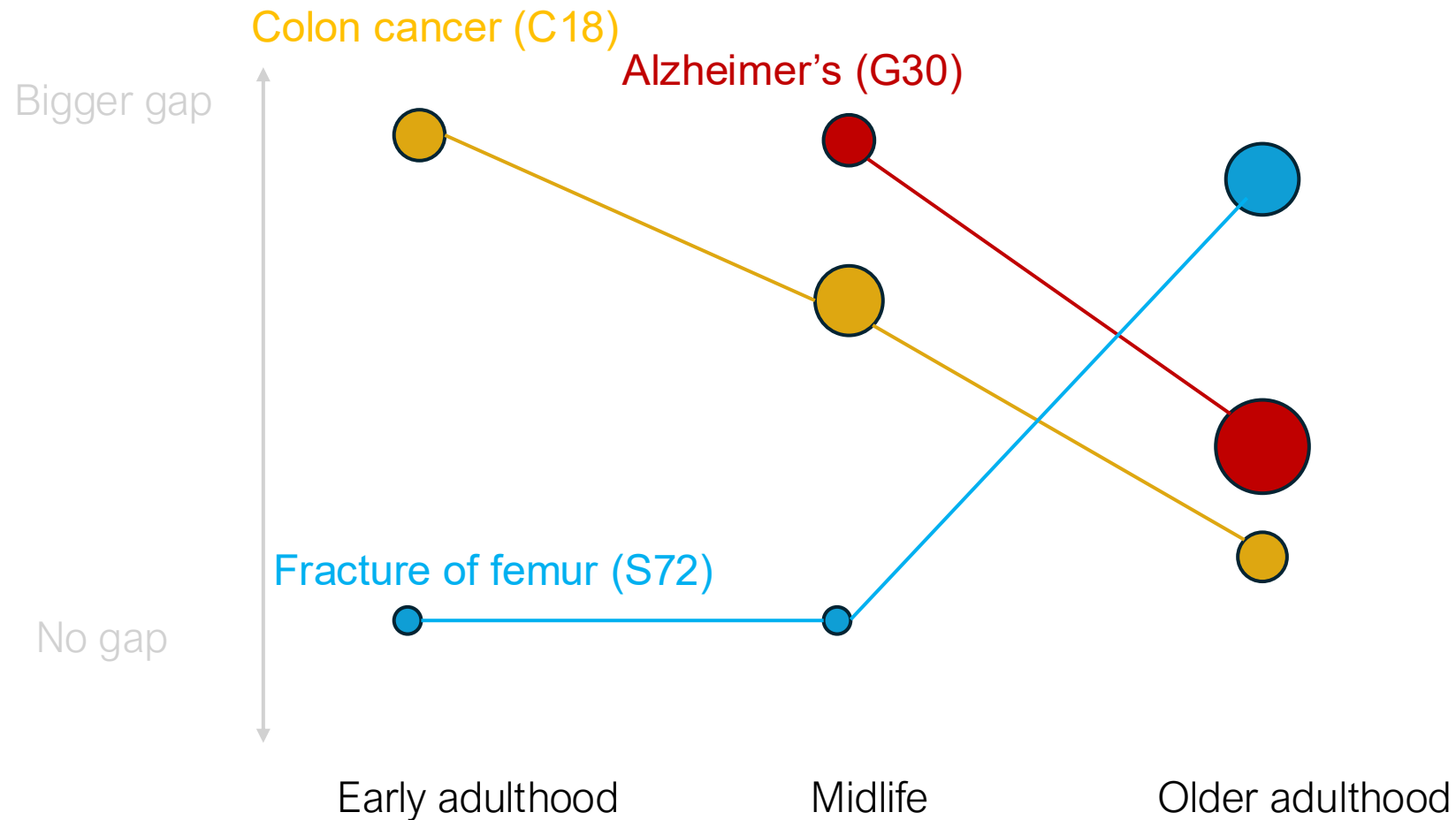
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But when we split by age...



Summary

- U.S. primary care has a distinct disease trajectory landscape: chronic and episodic clusters with identifiable bridge nodes
- Since musculoskeletal conditions are upstream, treating them earlier may be important
- Bridge nodes and turning points are orthogonal: population-level structural centrality does not predict trajectory-redirecting effects
- A counterfactual design can detect conditions that persistently redirect the downstream disease landscape (turning points), which vary by age
- More validation needed to see if the method can help generate new hypotheses about disease relationships and intervention points

Thank you



David Rehkopf
Stanford



Nathaniel Hendrix
Amer. Board Fam. Med.



Diego Guerrero
UC Santa Barbara



Laust Mortensen
Univ. of Copenhagen



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🦋 kspoon.bsky.social
✉️ kspoon@stanford.edu

So many thresholds to pick!

51% of pairs have fewer than 10 patients

76% have fewer than 50 patients

84% have fewer than 100 patients

$n \geq 100 + RR \geq 1.5$: 24,167 rows: across 10 bins that's ~2,400 edges per bin on average

$n \geq 500 + RR \geq 1.5$: 7,152 rows: ~715 edges per bin

$n \geq 1000 + RR \geq 1.5$: 3,830 rows: ~383 edges per bin

Inclusion criteria – full network

7,916,545 patients in system

5,774,940 patients have at least one visit (active)

3,586,684 patients have at least two visits, 6 months apart (62%)

The network has 10 time steps, one for each 6 month period.

Captures relative time, not absolute time.

Patients with visit span reaching END of each 6-month bin:

Bin	N patients	% of active

months 0-6	3,586,684	62.1%
months 6-12	3,070,024	53.2%
months 12-18	2,556,964	44.3%
months 18-24	2,145,565	37.2%
months 24-30	1,791,396	31.0%
months 30-36	1,504,426	26.1%
months 36-42	1,243,382	21.5%
months 42-48	1,023,085	17.7%
months 48-54	816,342	14.1%
months 54-60	625,973	10.8%

Community 0 (29 nodes):	Community 1 (5 nodes):	Community 5 (15 nodes):	Community 8 (32 nodes):
Acute infectious disease	Abnormal finding on evaluation procedure	Abdominal organ finding	Abnormal body temperature
Bacterial infectious disease	Visual system disorder	Abnormality of systemic vein	Acute inflammatory disease
Benign neoplasm of intra-abdominal organs	Neoplasm of intra-abdominal organs	Colonic lesion	Acute respiratory disease
Bladder finding	Neoplasm of uterus	Disease due to Gram-positive bacteria	Allergic disorder
Congenital disease	Primary malignant neoplasm of abdomen	Disorder of anal canal	Bacterial respiratory infection
Disease caused by parasite	Primary malignant neoplasm of trunk	Disorder of colon	Complication of pregnancy, childbirth and/or puerperium
Disorder of endocrine system	Community 2 (12 nodes):	Disorder of digestive organ	Degenerative disorder
Disorder of female genital system	Anomaly of eye	Disorder of intestine	Degenerative disorder of eye
Disorder of male genital organ	Complication	Disorder of large intestine	Delivery finding
Disorder of pelvic region of trunk	Complication due to diabetes mellitus	Disorder of pelvis	Disease due to Gram-positive coccus
Disorder of skin	Disorder due to type 2 diabetes mellitus	Disorder of the nose	Disorder due to infection
Disorder of skin appendage	Disorder of carbohydrate metabolism	Epithelial neoplasm of skin	Disorder of face
Disorder of the genitourinary system	Disorder of eye	High body weight	Disorder of fetus or newborn
Disorder of uterus	Infection by Ascomycetes	Injury of head	Disorder of lacrimal system
Evaluation finding	Measurement finding	Tumor stage finding	Disorder of upper digestive tract
Female reproductive finding	Peripheral nerve disease	Community 6 (9 nodes):	Disorder of upper respiratory system
Finding of abdominopelvic segment of trunk	Retinal disorder	Disorder of gastrointestinal tract	Finding of head region
Finding of pelvic region of trunk	Retinopathy due to diabetes mellitus	Disorder of pregnancy	Finding of mouth region
Finding of substance level	Sequela of disorder	Fetal finding	Functional finding of gastrointestinal tract
Finding relating to sexuality and sexual activity	Community 3 (11 nodes):	Finding of measures of pregnancy	Head finding
Inflammation of specific body systems	Breast lump	Finding related to pregnancy	Inflammation of specific body organs
Inflammatory dermatosis	Cognitive function finding	Hypertensive disorder	Inflammatory disorder of head
Lower urinary tract finding	Cytopenia	Maternal AND/OR fetal condition affecting labor AND/OR delivery	Inflammatory disorder of upper respiratory tract
Neoplasm of endocrine gland	Disorder of artery	Pelvic organ finding	Intestinal infectious disease
Pain of truncal structure	Disorder of trunk	Pregnancy, childbirth and puerperium finding	Lesion of skin and/or skin-associated mucous membrane
Testicular finding	Finding of region of thorax	Community 7 (2 nodes):	Pain in limb
Urogenital finding	Hemoglobin low	Congenital anomaly of head	Perinatal disorder of growth and/or development
Vagina finding	Mass of body structure	Disorder of ocular adnexa	Poisoning
Viral disease	Mass of trunk		Respiratory tract infection
	Neoplasm affecting hematopoietic structure		Upper respiratory infection
	Primary malignant neoplasm		Viral respiratory infection
			Vital signs finding

Community 9 (17 nodes):
Arterial finding
Chest injury
Closed fracture
Closed fracture of lower limb
Closed fracture of upper limb
Disorder of body wall
Disorder of cardiac function
Disorder of mediastinum
Disorder of shoulder
Disorder of thigh
Disorder of thorax
Disorder of vertebral column
Fracture of bone
Injury of free lower limb
Injury of musculoskeletal system
Injury of trunk
Vascular disease of abdomen

Community 10 (33 nodes):
Arthropod dermatosis
Disorder of digit
Disorder of extremity
Disorder of finger
Disorder of foot
Disorder of free lower limb
Disorder of hand
Disorder of integument
Disorder of joint region
Disorder of lower extremity
Disorder of soft tissue
Disorder of soft tissue of head
Disorder of soft tissue of limb
Disorder of soft tissue of lower limb
Disorder of upper extremity
Finding of back
Finding of hand region
Finding of limb structure
Finding of upper limb
General clinical state finding
Inflammation of specific body structures or tissue
Injury of upper extremity
Malignant neoplastic disease
Musculoskeletal finding
Neoplasm of skin
Neoplastic disease
Pain
Skin lesion
Soft tissue lesion
Superficial injury
Traumatic injury
Traumatic or non-traumatic injury
Wound

Community 11 (2 nodes):
Disorder of lower respiratory system
Disorder of lung

Community 12 (3 nodes):
Blindness AND/OR vision impairment level
Disorder of vision

Community 4 (101 nodes):

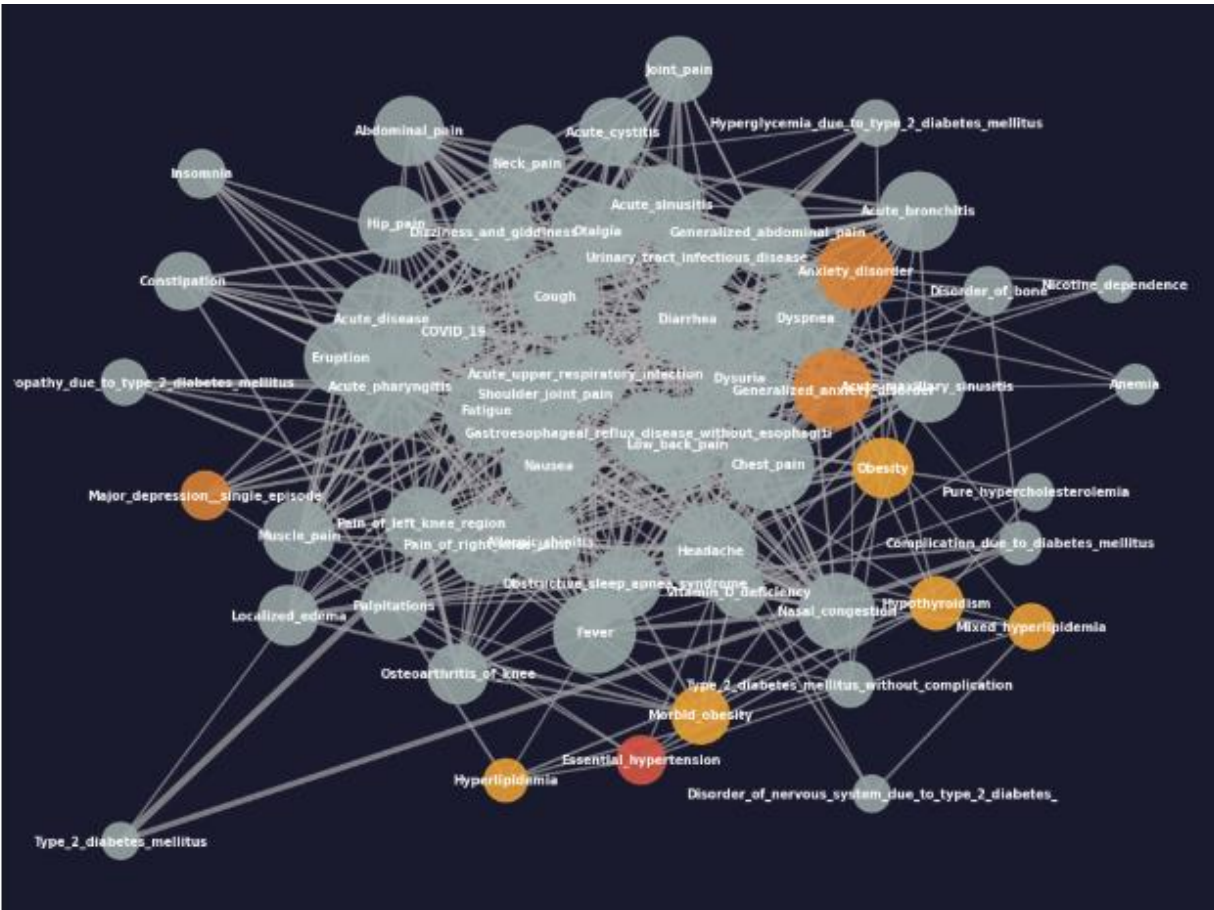
Allergic disorder of respiratory tract
Anemia
Anxiety
Arthropathy
Atrial cardiopathy
Behavior finding
Bipolar disorder
Cardiac arrhythmia
Cardiac finding
Cardiovascular finding
Cardiovascular measurement - finding
Chronic disease of respiratory system
Deficiency of micronutrients
Depressive disorder
Developmental disorder
Developmental mental disorder
Disease due to Gram-negative bacteria
Disease due to Paramyxoviridae
Disease related state
Disorder of abdomen
Disorder of abdominopelvic segment of trunk
Disorder of amino acid metabolism
Disorder of auditory system
Disorder of bone
Disorder of brain
Disorder of cardiac ventricle
Disorder of cardiovascular system
Disorder of cellular component of blood
Disorder of connective tissue
Disorder of ear
Disorder of epiphysis
Disorder of external ear
Disorder of head
Disorder of immune function
Disorder of joint of spine

Disorder of knee
Disorder of lipoprotein AND/OR lipid metabolism
Disorder of lipoprotein storage and metabolism
Disorder of mineral metabolism
Disorder of musculoskeletal system
Disorder of neck
Disorder of nervous system
Disorder of prostate
Disorder of psychological development
Disorder of respiratory system
Disorder of retroperitoneum
Disorder of right ear
Disorder of skeletal system
Disorder of spinal region
Disorder of stomach
Disorder of the central nervous system
Disturbance of consciousness
Drug dependence
Drug-related disorder
Dyssomnia
Ear, nose and throat disorder
Emotional state finding
Finding of functional performance and activity
Finding of respiration
Finding of sensation by site
Finding of spinal region
Finding of thought content
Finding related to ability to move
General finding of soft tissue
Hearing disorder
Heart block
Heart disease
Heart failure
Hyperlipidemia
Inflammatory disorder of extremity
Inflammatory disorder of the respiratory tract

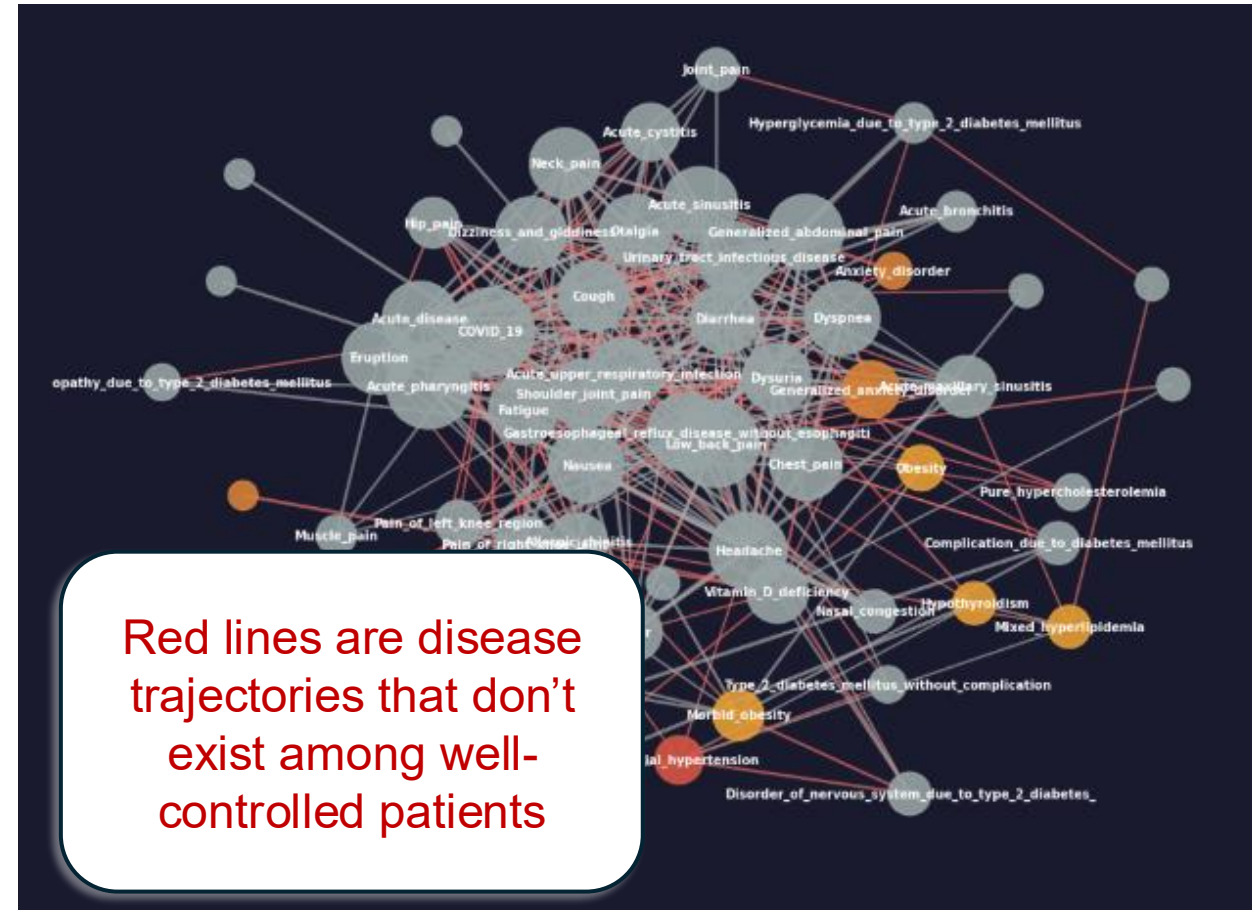
Joint finding
Kidney disease
Knee joint finding
Language finding
Lesion of joint
Level of consciousness - finding
Measurement finding above reference range
Mental disorder
Mental state finding
Mental state, behavior and/or psychosocial function finding
Metabolic disease
Mood disorder
Musculoskeletal and connective tissue disorder
Mycosis
Necrosis of anatomical site
Neurological lesion
Pain / sensation finding
Respiratory finding
Seizure disorder
Sensory nervous system finding
Sequela
Speech finding
Sprain of joint
Structural disorder of heart
Tracheobronchial disorder
Traumatic or nontraumatic brain injury
Undernutrition
Urinary system finding
Vascular disorder
Viscus structure finding

The disease landscape is **structurally different** between groups

Well controlled diabetes ($HbA1c < 7$)



Poorly controlled diabetes ($HbA1c > 9$)



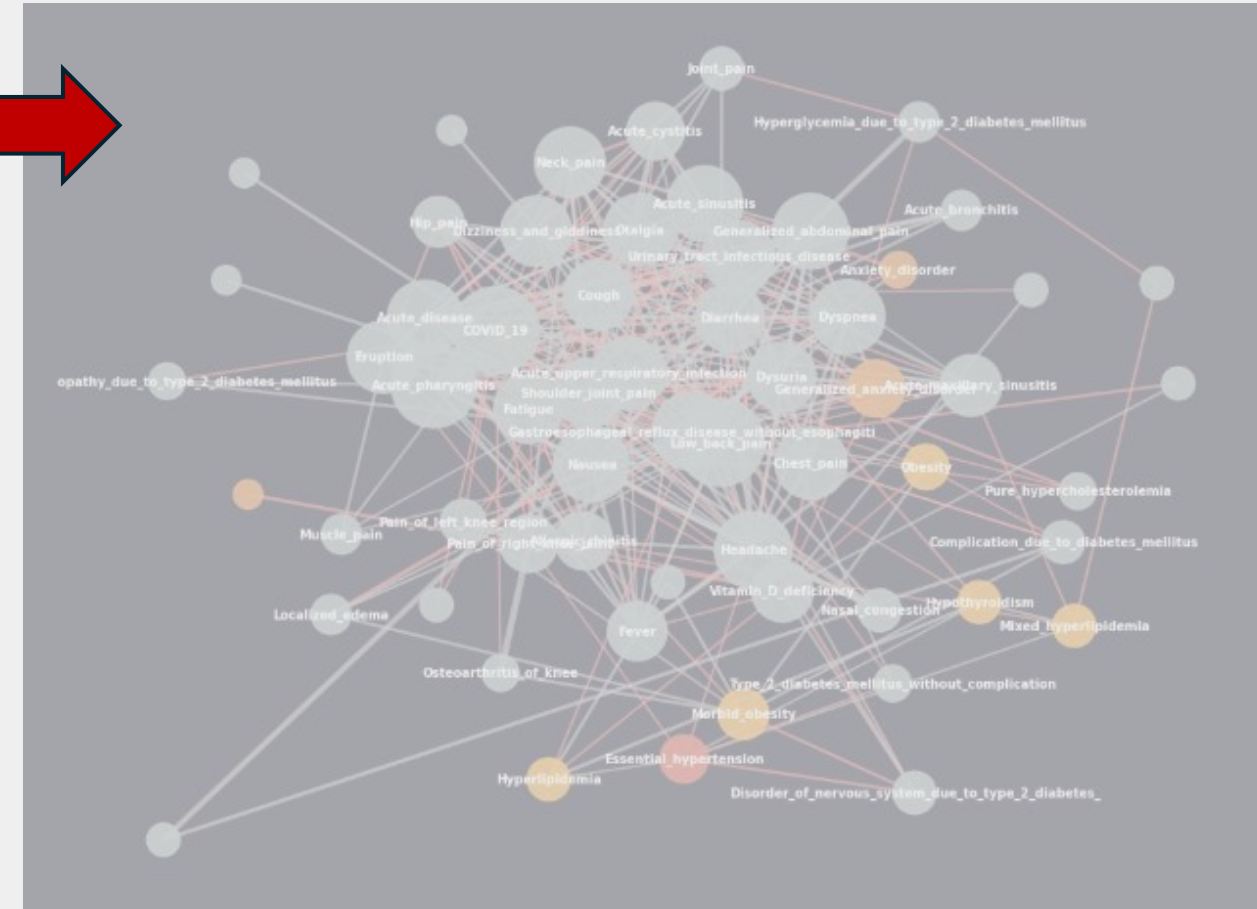
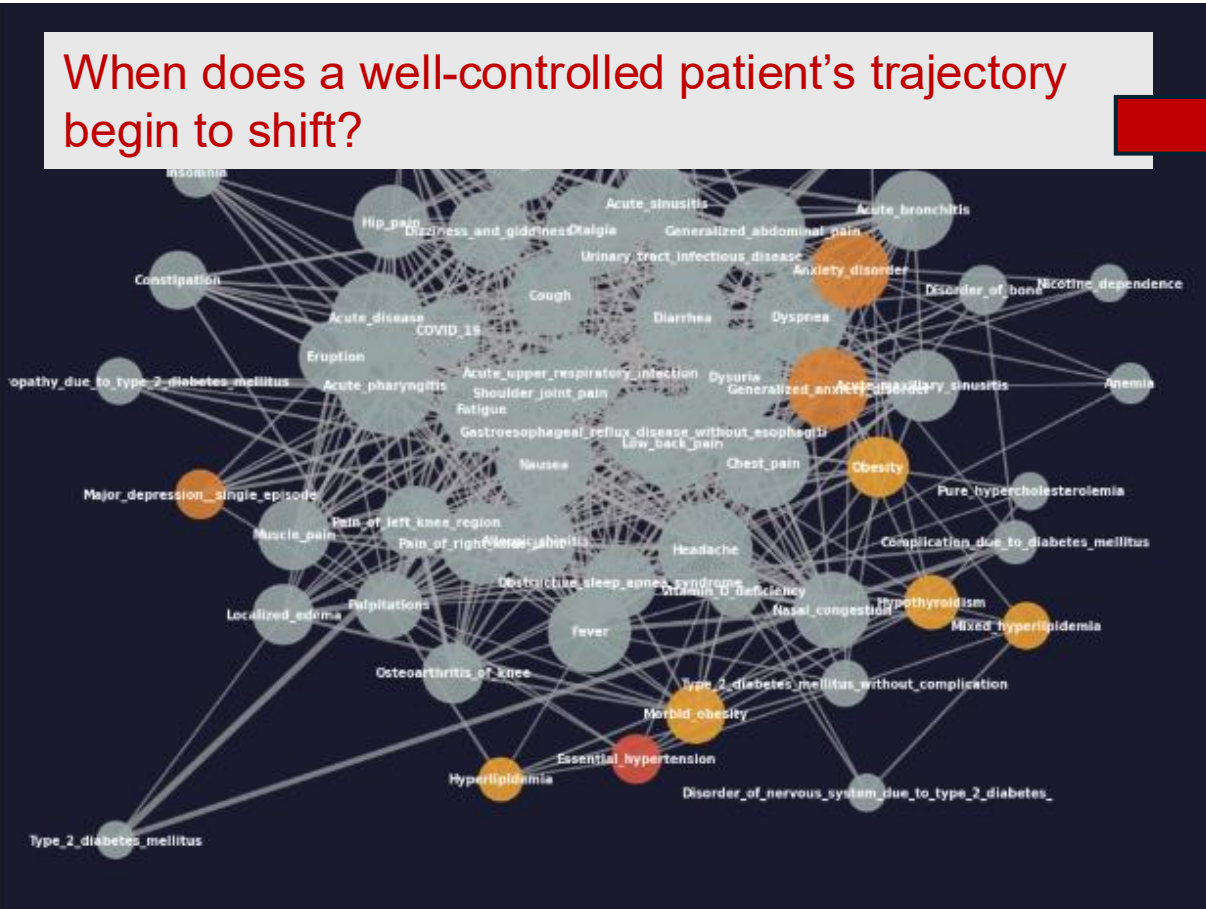
The disease landscape is **structurally different** between groups

Well controlled diabetes (Hba1c < 7)

When does a well-controlled patient's trajectory begin to shift?

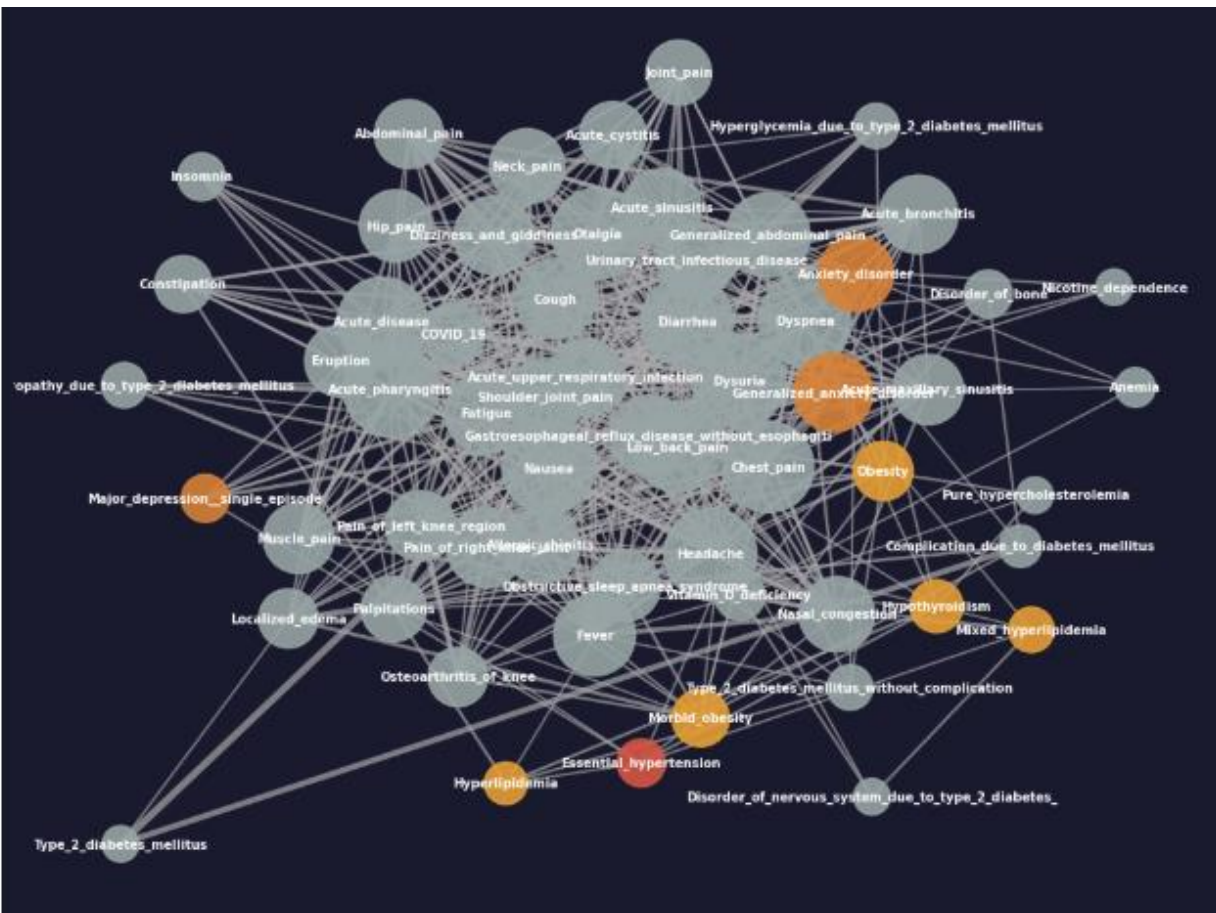


Poorly controlled diabetes (Hba1c > 9)



Depression is a possible turning point

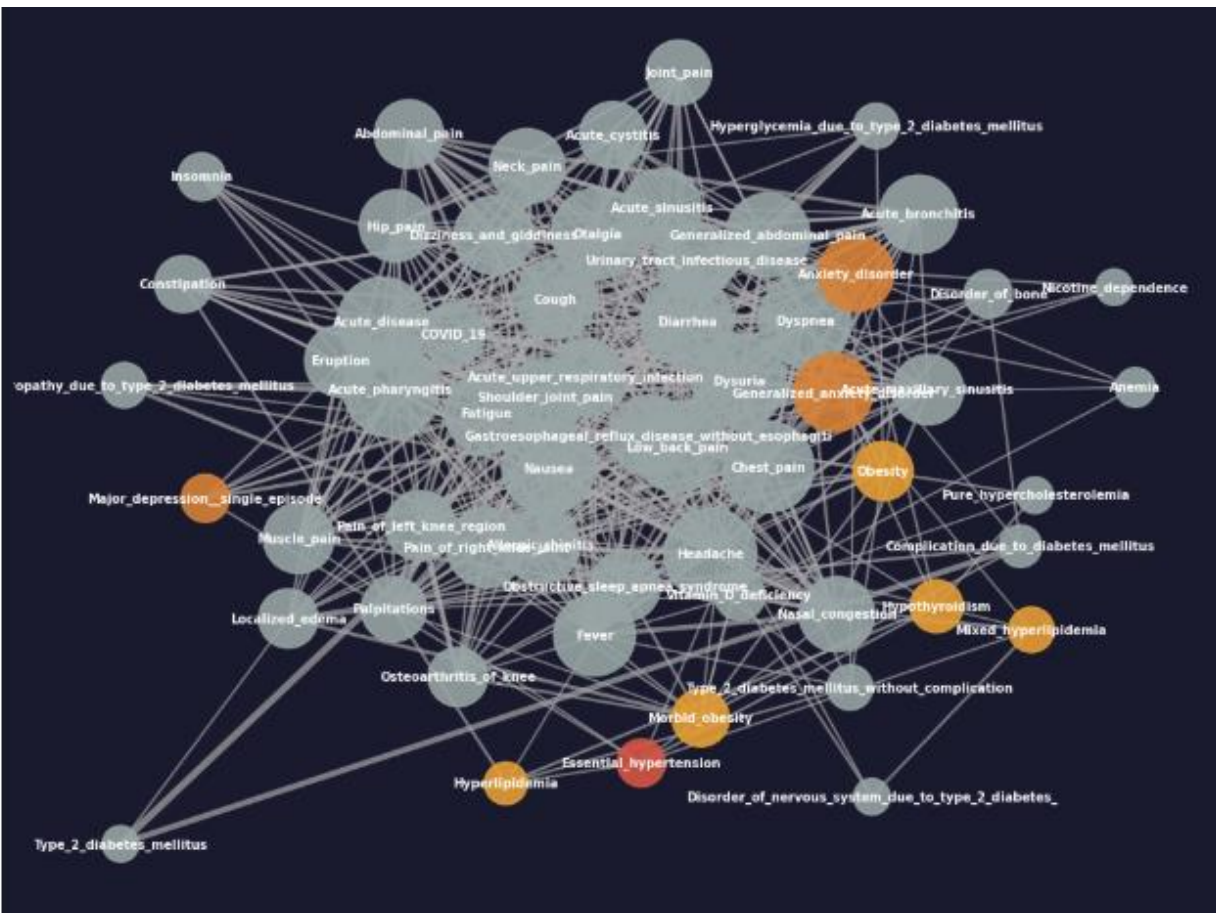
Well controlled diabetes ($HbA1c < 7$)



Well-controlled patients diagnosed with **depression** have **more chronic conditions** later, than those not diagnosed with depression

Depression is a possible turning point

Well controlled diabetes ($HbA1c < 7$)



Well-controlled patients diagnosed with **depression** have **more chronic conditions** later, than those not diagnosed with depression

If we can identify turning points, we can intervene before complications compound

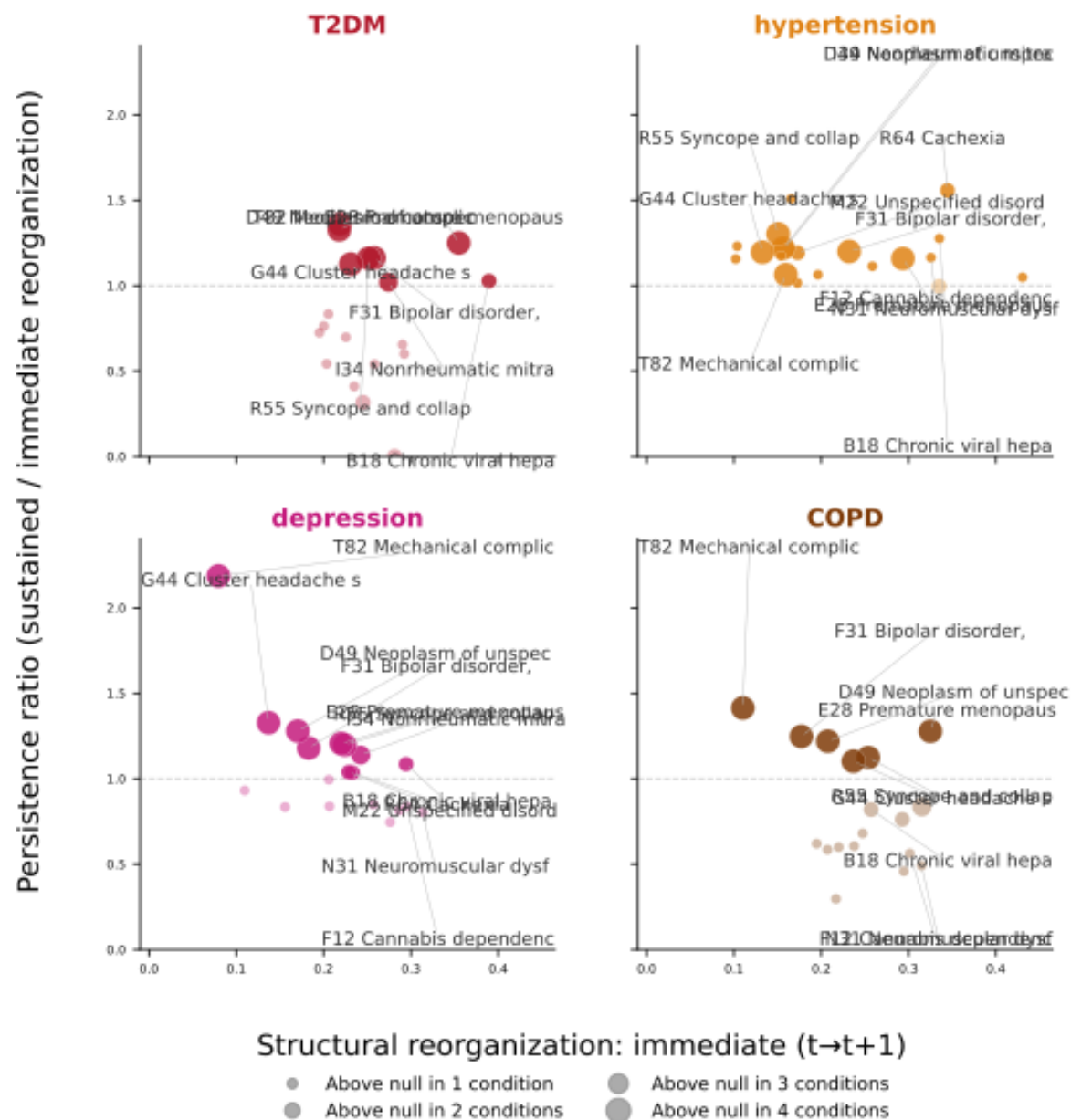
Turning points

A condition is a turning point if acquiring it:

1. Restructures the downstream disease network (edge rewiring)
2. Does so **persistently** across time

Tested via matched acquirer / non-acquirer comparison

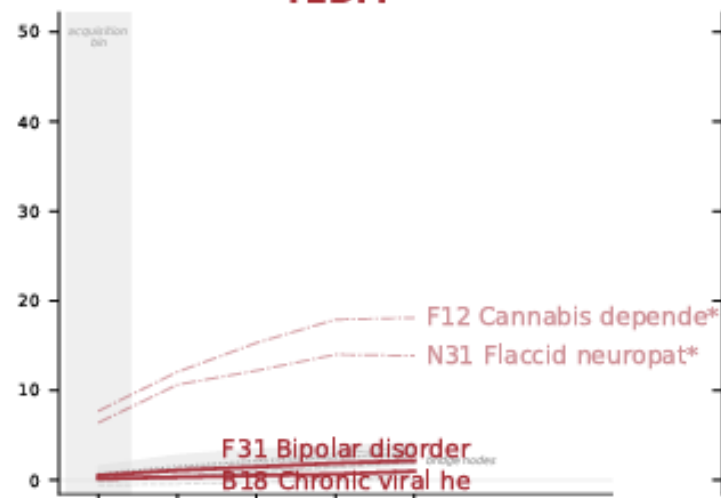
Does that effect
strengthen or
attenuate over
time?



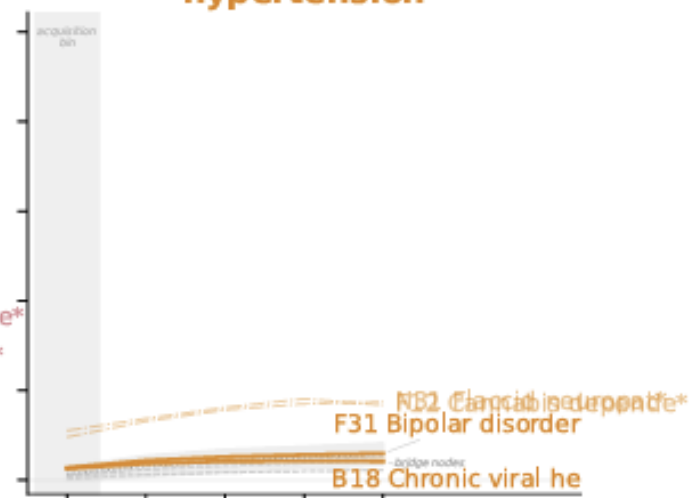
To what degree does a condition rewire the network? (% edges changed)

Cumulative gap in new conditions
(acquirers – matched non-acquirers)

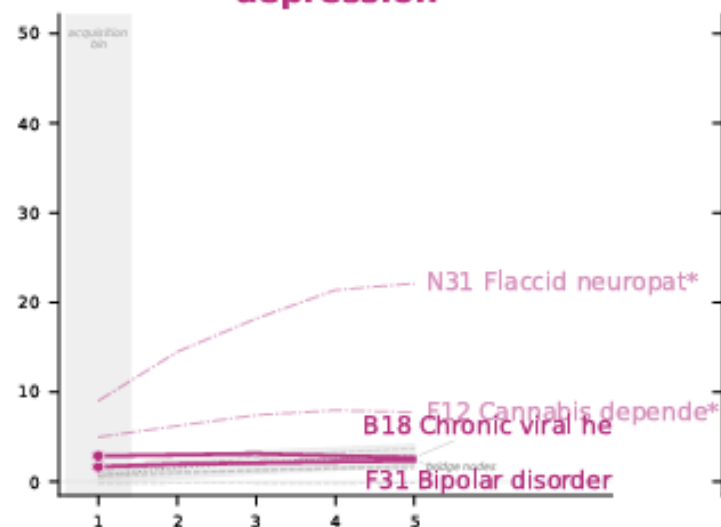
T2DM



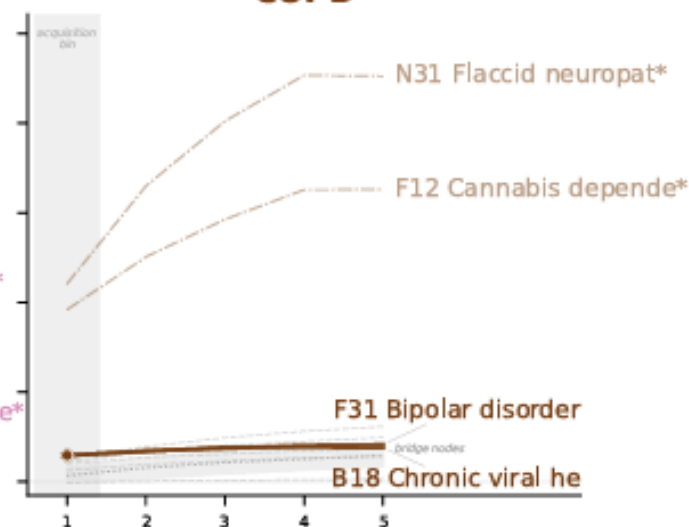
hypertension



depression



COPD

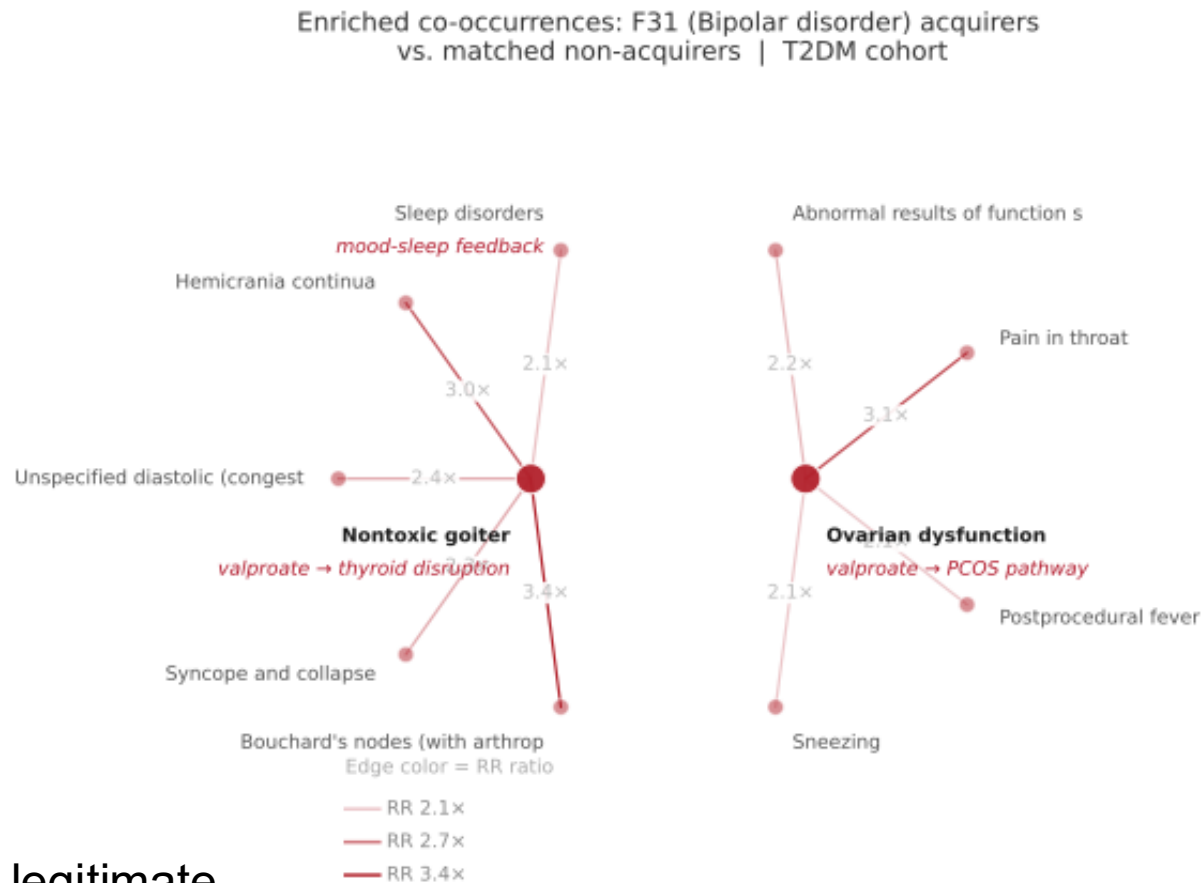


Bins post-acquisition (6-month intervals)

— Turning point candidates - - - Severity markers . . . Bridge nodes

Recovering a known turning point

All of these connections exist only among those who acquired bipolar disorder, which opened a whole new part of the network



It seems like it is recovering legitimate turning points (known from valproate literature)

Bridge nodes $\neq$ turning points

- Bridge nodes (high betweenness centrality in the population network) do not appear as turning point candidates
- Structural centrality in the population-level network and trajectory-redirecting effects at the individual level are orthogonal
- Turning points must be detected dynamically: static network structure is not sufficient